Bayesian Surface Habitability Mapping of Titan

from the Late Heavy Bombardment to the Solar Red-Giant Phase

A report submitted for the degree of

Master of Philosophy

Planetary Science and Life in the Universe

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All code, processed data products, and static habitability maps are openly archived at https://github.com/chrimblechrumble/pslu-p3.

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1 Introduction

1.1 Titan as an Astrobiological Target

The search for habitable environments beyond Earth has increasingly focused on icy ocean worlds in the outer solar system. Among these, Saturn's moon Titan occupies a singular position. It is the only moon with a dense atmosphere — dominated by molecular nitrogen at 1.467 bar surface pressure and 93.65 K (Fulchignoni et al., 2005) (Appendix A) — and the only known extraterrestrial world to host stable surface liquids (Stofan et al., 2007), completing a full precipitation–evaporation–lake cycle structurally analogous to Earth's hydrological cycle. Its atmosphere continuously produces complex organic tholins at $\sim 5 \times 10^{-14}\,\mathrm{g\,cm^{-2}\,s^{-1}}$ (Lavvas et al., 2011), building a global organic sediment layer over ~4 Gyr (Lorenz et al., 2008). At depth, a global water–ammonia ocean at ~55–80 km is confirmed by the tidal Love number $k_2 = 0.589 \pm 0.150$ (Iess et al., 2012). Tidal dissipation from Saturn's gravitational forcing maintains the ocean's liquid state and may generate fractures or brine conduits through the ice shell which is researched in (Crick, 2026). This convergence of cycling solvent, vast organic inventory, and subsurface water makes Titan a natural laboratory for prebiotic chemistry operating at planetary scale across geological time.

1.2 The Physical Environment

The atmosphere (~94% N_2, 5.65% CH_4; Niemann et al. 2005) sustains a methane cycle with observed equatorial rainfall (Turtle et al., 2011), fluvial channel networks (Miller et al., 2021), and polar hydrocarbon seas (Kraken, Ligeia, and Punga Mare) containing ~70,000 $\mathrm{km^3}$ of liquid methane–ethane (Malaska et al., 2025; Mastrogiuseppe et al., 2019). The lake distribution is strikingly asymmetric: the north polar region hosts hundreds of filled lakes and seas covering ~1.6% of the global surface, while equivalent southern latitudes are largely devoid of standing liquid (Aharonson et al., 2009). This hemispheric imbalance arises from the eccentricity of Saturn's orbit, which makes northern summers longer and milder than southern summers, driving net methane transport from south to north over Milankovitch-like cycles of ~45,000 yr (Aharonson et al., 2009). The asymmetry is directly relevant to the habitability maps: present-epoch solvent habitability is concentrated in the liquid-rich north, but the orbital forcing implies that the southern hemisphere hosted comparable lake systems in the geologically recent past and will do so again. The surface is geologically diverse: equatorial dune fields (~17% by SAR area), plains (~65%), labyrinth terrain, impact craters, and mountains, as classified by the Lopes et al. (2019b) global geomorphological map that provides the terrain classification underpinning this thesis.

1.3 Defining Habitability

Habitability is a term used in diverse and sometimes incompatible ways across the planetary science literature (Cockell et al., 2016). In its strictest sense it denotes a binary property — whether an environment can sustain the metabolic activity of at least one known organism — but in practice astrobiologists use it as a continuous measure of how many prerequisites for life co-occur in a given location (Méndez et al., 2021). This thesis adopts the continuous usage. The posterior probability $P(H \mid \mathbf{f})$ at each pixel quantifies the degree to which the *necessary physical and chemical preconditions* for life are simultaneously present: liquid solvent, organic substrate, chemical energy, solvent cycling, atmosphere–surface exchange, topographic and geomorphological diversity, and subsurface water access. High $P(H)$ does not imply that life exists at a site, nor that abiogenesis could occur there; it implies that more of the known prerequisites are satisfied, and satisfied more strongly, than at a low-$P(H)$ site.

Abiogenesis — the emergence of life from non-living chemistry — demands conditions beyond those captured by the habitability proxy. In aqueous systems, the minimum requirements include liquid water, a source of organic monomers, free energy to drive polymerisation, and a mechanism for compartmentalisation (Ruiz-Mirazo et al., 2014). On Titan, two fundamentally different abiogenesis scenarios are conceivable: *aqueous abiogenesis* in transient impact-melt ponds or the future red-giant ocean, where liquid water contacts accumulated organics; and *non-aqueous abiogenesis* in the present-day hydrocarbon lakes, where azotosome membranes (Stevenson et al., 2015) could provide compartmentalisation and acetylenotrophy (McKay and Smith, 2005) could provide energy, but where the low temperature (94 K) severely constrains reaction kinetics. The habitability maps produced in this thesis therefore represent a *necessary but not sufficient* condition for either form of abiogenesis: they identify where the preconditions converge spatially and temporally, but cannot assess whether the additional requirements for the origin of life — sustained chemical disequilibrium, information-carrying polymers, or encapsulation — are met.

1.4 Chemical Habitability Scenarios

Four habitability pathways have been proposed for Titan's surface, each with distinct implications for abiogenesis.

1. *Methanogenesis via acetylenotrophy* (McKay and Smith, 2005; Yanez et al., 2024): the hydrogenation of acetylene, $C_2H_2 + 3\,H_2 \longrightarrow 2\,CH_4$ ($\Delta G = -334\,\mathrm{kJ\,mol^{-1}}$), produces methane as the metabolic end product and provides energy well above the minimum for terrestrial methanogens. McKay and Smith (2005) proposed this as a basis for methanogenic life on Titan; Yanez et al. (2024) refined the thermochemistry under Titan ocean conditions, finding acetylenotrophy energy yields of 69–78 $\mathrm{kJ\,mol^{-1}\,C}$. Acetylene is supplied by UV photolysis and detected at the surface by GCMS (Niemann et al., 2005). This pathway could sustain an existing non-aqueous biosphere but does not by itself provide a mechanism for abiogenesis.

2. *Azotosome membranes* (Stevenson et al., 2015): acrylonitrile (detected by ALMA; Palmer et al. 2017) spontaneously forms stable vesicle structures in liquid methane, providing a route to cellular compartmentalisation without aqueous chemistry. Unlike the fatty acid vesicles or coacervate droplets proposed for aqueous abiogenesis (Ruiz-Mirazo et al., 2014), azotosomes represent a fundamentally different compartmentalisation solution operating in a non-polar solvent. Compartmentalisation is a key requirement for abiogenesis; azotosomes could in principle fulfil this role in a non-aqueous origin-of-life scenario, though no complete pathway from vesicle to replicator has been demonstrated.

3. *Impact-melt prebiotic chemistry*: large impacts produce transient water–ammonia–organic melt ponds persisting for 10^3–10^4 yr, during which Strecker synthesis yields amino acids from surface aldehydes, ammonia, and HCN (Madan et al., 2026; Neish et al., 2010). Of all present-day scenarios, this provides the most direct route to aqueous abiogenesis, since liquid water, organic monomers, and energy are simultaneously available, though the transient duration ($\sim 10^4$ yr) may be too short to complete the transition from prebiotic chemistry to self-replicating systems. See (Nicolaides, 2026) for crater physics modelling quantitative melt-volume research. See (Castlevetro, 2026) for research relating to abiogenesis within ammonia-rich brine pockets within Titans icy crust. See (Rahim, 2026) regarding the exogenicendogenic mixing of surface organics with excavated water-ice at crater sites and its relevance to the prebiotic chemistry.

4. *Red-giant ocean*: when solar luminosity raises Titan's surface above the water–ammonia eutectic (176 K; Grasset and Pargamin 2005), a global ocean will form over billions of years of accumulated organics for an estimated 200–500 Myr (Lorenz et al., 1997). This is the most favourable abiogenesis scenario modelled in this thesis: warm liquid water in sustained contact with $\sim$10 Gyr of accumulated organic substrate, persisting for hundreds of millions of years — conditions broadly analogous to those invoked for early Earth.

This thesis provides the first unified quantitative characterisation of all four scenarios across geological time.

1.5 Scope and Research Questions

Previous habitability assessments of Titan have been predominantly qualitative or point-specific: the Earth Similarity Index (Schulze-Makuch et al., 2011) produces a single scalar per body with no spatial resolution; Hendrix et al. (2019) reviewed icy ocean world habitability using qualitative criteria matrices; Affholder et al. (2021) applied rigorous Bayesian inference to Enceladus plume data but at a single spatial point and epoch. None integrates the full suite of Cassini data modalities into a spatially-resolved, multi-epoch probabilistic framework. A detailed methodological comparison is given in Appendix J.

This thesis develops a Bayesian Beta-conjugate posterior framework combining eight observational proxy features from the complete Cassini dataset into a habitability probability map at 4490 m/px resolution, applied across five temporal epochs from the Late Heavy Bombardment (-3.5 Gya) to the red-giant future ($+5.9$ Gya). The Beta-conjugate approach is preferred for its analytical tractability at 6.49 million grid pixels, its naturally bounded $[0, 1]$ output domain, and the explicit uncertainty quantification provided by 95% highest-density interval credible intervals at every pixel.

The specific research questions are:

RQ1 *What is the spatial distribution of present-day surface habitability, and which environments produce the highest scores?*

RQ2 *How did habitability differ during the Late Heavy Bombardment, when impact-melt ponds provided transient liquid water?*

RQ3 *How does habitability evolve across the temporal sequence from the LHB (-3.5 Gya) to the red-giant ocean ($+5.9$ Gya)?*

RQ4 *Does the derived habitability map independently validate the Dragonfly landing site at Selk crater?*

2 Methods

2.1 Overview of the Analytical Approach

The framework proceeds in four stages: (1) extraction of eight habitability-proxy features from the Cassini observational archive (Section 2.3); (2) Bayesian conjugate updating to produce a posterior habitability probability map at each epoch (Section 2.4); (3) temporal scaling of features across five anchor epochs and 72 intermediate geological epochs from -3.8 Gya to $+6.5$ Gya (Sections 2.5–2.6); and (4) ranking of named surface sites by posterior probability. The posterior is evaluated at every pixel of the canonical equirectangular grid, producing global maps at each epoch from which habitable-zone boundaries, site rankings, and temporal trajectories can be extracted. Acronyms, notation, and the full data-source catalogue are defined in Appendix F.

The analysis is applied at 4490 m/px resolution on a canonical equirectangular grid covering the full globe (1802 pixels in latitude $\times$ 3603 pixels in longitude $= 6.49\times10^6$ pixels), using a west-positive longitude convention consistent with all Cassini raster products. Raw GeoTIFF data products are reprojected onto this grid using `rasterio.warp.reproject` with bilinear resampling; polar inset maps use an oblique stereographic projection (Snyder, 1987) (Appendix F.4). The grid resolution corresponds to $\sim$20 km at the equator, comparable to the spatial scale of terrain-class boundaries in the Lopes et al. (2019b) geomorphological map and sufficient to resolve individual lake margins, crater structures, and dune-field boundaries. Figure 2.1 provides a schematic overview.

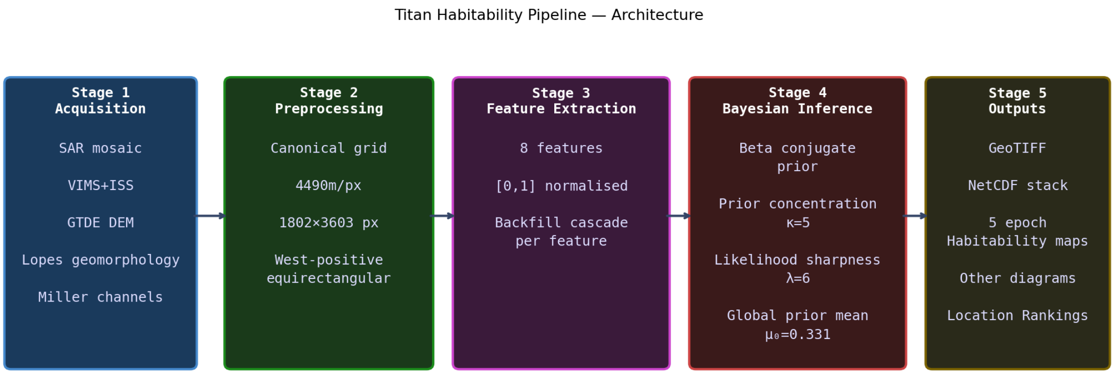

Figure 2.1: Schematic overview of the five-stage Titan habitability analysis pipeline. All eight habitability-proxy features are extracted in Stage 3 and combined in the Bayesian posterior update of Stage 4 to produce $P(H \mid \mathbf{f})$ maps at each of the five temporal configurations.

2.2 Observational Data Sources

Eight Cassini data products spanning SAR imaging, near-infrared spectroscopy, thermal emission, altimetry, geomorphological classification, fluvial channel mapping, gravity, and crater cataloguing are used (Table 2.1; Appendix H provides an overview of the Cassini–Huygens and *Dragonfly* missions). The primary spatial data sources are the SAR backscatter mosaic (∼67% coverage at ∼350 m/px), the VIMS+ISS 1.59/1.27 µm band-ratio mosaic (∼50% coverage; Seignovert et al. 2019), and the Lopes et al. (2019b) global geomorphological classification. Processing details are in Appendix F.

Table 2.1: Cassini data sources used in the habitability framework.

Dataset	**Coverage**	**Habitability feature**
Cassini SAR mosaic	∼67% surface	f_1 liquid HC; f_7 geodiversity
VIMS 1.59/1.27 µm	∼50% surface	f_2 organic abundance
CIRS thermal mosaic	Global (∼10°)	f_4 methane cycle; f_5 surf–atm
GTDE/GTDR DEM	∼90% (interp.)	f_6 topographic complexity
Lopes (2019) geomorph. map	∼90% SAR	f_2 organic; f_7 geodiversity
Miller (2021) channel map	SAR coverage	f_5 surface–atmosphere
Tidal k_2 (global)	Global scalar	f_8 subsurface ocean
Hedgepeth (2020) craters	SAR coverage	f_7 geodiversity; past epoch

2.3 Feature Extraction

Eight habitability-proxy features f_1–f_8 are extracted from the data sources above, each normalised to $[0, 1]$ via robust percentile clipping (2nd–98th percentile; Eq. 2.1). The features are designed to capture the principal habitability-relevant physical conditions on Titan's surface: the availability of liquid solvent (f_1), the inventory of prebiotic organic substrate (f_2), the energy supply for metabolic reactions (f_3), the intensity of solvent cycling that drives chemical transport and concentration (f_4), the strength of atmosphere–surface chemical exchange (f_5), the topographic diversity that creates distinct micro-environments (f_6), the terrain-class heterogeneity that proxy for geochemical interfaces (f_7), and the proximity to the subsurface water–ammonia ocean (f_8). The weights w_i (Table 2.2) reflect the relative astrobiological importance of each condition, assigned from physical reasoning and independently validated against data-derived feature importances (Section 3.4.3).

$$\hat{f}_i = \mathrm{clip}\left(\frac{f_i - f_{i,\mathrm{p2}}}{f_{i,\mathrm{p98}} - f_{i,\mathrm{p2}}},\ 0,\ 1\right) \tag{2.1}$$

Table 2.2: The eight habitability-proxy features. Weight w_i and prior mean μ_i are the present-epoch values used in the Bayesian update (Eq. 2.11). $\mu_0 = \sum_i w_i \mu_i = 0.331$.

f_i	**Name**	w_i	μ_i	**Scientific rationale**
f_1	Liquid HC availability	0.25	0.02	Direct solvent for non-aqueous biochemistry (McKay and Smith, 2005)
f_2	Organic abundance	0.20	0.70	Tholin substrate for prebiotic chemistry (Meyer-Dombard et al., 2025)
f_3	Chemical energy (acetylene)	0.20	0.35	Acetylenotrophy energy proxy (McKay and Smith, 2005; Yanez et al., 2024)
f_4	Methane cycle intensity	0.15	0.40	Solvent transport and chemical concentration cycling (Mitchell and Lora, 2016; Turtle et al., 2011)
f_5	Surface–atm. interaction	0.08	0.35	Organic delivery at lake margins and channel termini (Hayes, 2016; Miller et al., 2021)
f_6	Topographic complexity	0.06	0.25	Microenvironment diversity; roughness exposes water-ice crust (Lopes et al., 2019b)
f_7	Geomorphological diversity	0.04	0.30	Chemical gradients at terrain-type interfaces (Lopes et al., 2019b; Shock and Holland, 2007)
f_8	Subsurface ocean proximity	0.02	0.03	Potential water-chemistry conduit; SAR-bright annuli (Iess et al., 2012)

Figure G.1 (Appendix G) shows all eight feature maps at the present epoch, illustrating their distinct spatial patterns: the polar concentration of liquid hydrocarbon (f_1), the near-global organic abundance (f_2), the topographically modulated chemical energy proxy (f_3), and the latitude-dependent methane cycle signal (f_4). Figure G.2 shows how the relative activity of each feature is modulated across geological time. Implementation details, backfill strategies, and equations for composite features are given in Appendix G; the scientific basis for each feature is summarised below.

2.3.1 Feature Descriptions

f_1: Liquid hydrocarbon availability ($w_1 = 0.25$, $\mu_1 = 0.02$). Liquid methane–ethane is the proposed non-aqueous biochemical solvent for any surface life on Titan (McKay, 2016; McKay and Smith, 2005). At 94 K, water is structurally rigid and cannot support the molecular diffusion processes fundamental to metabolism; liquid methane and ethane are stable under Titan's surface conditions, as confirmed by the permanent north polar seas. The feature is derived from the Cassini SAR lake mask. The prior mean $\mu_1 = 0.02$ reflects the ∼1.5–2% global areal coverage of confirmed lakes and seas (Hayes et al., 2008). It receives the highest weight because solvent availability is a prerequisite for any biochemistry; despite this, the low prior mean means that most pixels contribute negligibly through this feature, and its habitability impact is strongly localised to the polar lake margins.

f_2: Organic abundance ($w_2 = 0.20$, $\mu_2 = 0.70$). Tholins produced by UV photolysis of the N_2/CH_4 atmosphere are the dominant surface material across most of Titan and the primary chemical substrate for any habitability scenario (Meyer-Dombard et al., 2025). Laboratory hydrolysis of tholins in liquid water produces amino acids and nucleobase analogues at ~0.1% yield (Neish et al., 2018), confirming their prebiotic potential. The feature is derived from the VIMS 1.59/1.27 µm band ratio where coverage exists (~50% of the globe), and from the Lopes et al. (2019b) geomorphological terrain class scores elsewhere (Appendix D; providing seamless global coverage without the spectral seam that arises from combining VIMS and ISS data; Section G.2). The prior mean $\mu_2 = 0.70$ reflects that ~82% of the surface falls into organic-rich terrain classes (dunes 17% + plains 65%; Malaska et al. 2025).

f_3: Chemical energy — acetylene ($w_3 = 0.20$, $\mu_3 = 0.35$). Life requires sustained chemical free energy. The primary candidate on Titan is the hydrogenation of acetylene ($\Delta G = -334\,\mathrm{kJ\,mol^{-1}}$; McKay and Smith 2005), with updated energy yields of 69–78 $\mathrm{kJ\,mol^{-1}}$ C under Titan ocean conditions (Yanez et al., 2024). Since no Cassini instrument directly maps surface acetylene concentration, the feature is approximated from SAR backscatter (dark surfaces indicate organic-rich terrain where acetylene accumulates; 70% weight) and topographic inversion (atmospheric condensates settle in topographic lows; 30% weight). The prior mean $\mu_3 = 0.35$ reflects the global availability of acetylene from continuous atmospheric photolysis, tempered by the indirect nature of the proxy.

f_4: Methane cycle intensity ($w_4 = 0.15$, $\mu_4 = 0.40$). The methane cycle is the dominant agent of chemical transport on Titan, delivering organic material to polar lake margins, maintaining chemical gradients at shorelines, and providing seasonal wet–dry cycling that may concentrate organic reactants (Mitchell and Lora, 2016; Turtle et al., 2011). The feature is a weighted composite of three independently computed components: VIMS observation density (50%; observation targeting correlates with active methane-cycle regions), the meridional temperature gradient from CIRS (Jennings et al., 2019) (25%; steep gradients drive evaporation differentials), and a Gaussian latitude prior centred at ±45° (25%; polar regions where the methane cycle is most active). The prior mean $\mu_4 = 0.40$ reflects the broad spatial extent of the active methane cycle across mid-to-high latitudes.

f_5: Surface–atmosphere interaction ($w_5 = 0.08$, $\mu_5 = 0.35$). The physical interface between surface and atmosphere is where gas-phase photochemical products encounter liquid solvents and organic substrates. Lake margins and fluvial channel termini are the primary loci of this exchange (Hayes, 2016; Miller et al., 2021). The feature combines topographic slope (30%; steep slopes drive physical transport), a lake-margin proximity mask (40%; evaporation zones), and the Miller et al. (2021) global channel density map (30%). The lower weight ($w_5 = 0.08$) reflects that this feature captures a secondary process — the mechanism of organic delivery rather than its chemical significance.

f_6: Topographic complexity ($w_6 = 0.06$, $\mu_6 = 0.25$). Local topographic roughness at ~20 km scale correlates with dissected terrain that most frequently exposes water-ice-bearing crustal material (Lopes et al., 2019b). On Earth, topographic heterogeneity creates micro-environmental diversity that underpins elevated species richness at landscape boundaries; on Titan, the analogous principle is that varied topography produces diverse chemical micro-environments. The feature is the local standard deviation of GTDE elevation in a 5×5 pixel ($\sim$22 km) sliding window. The prior mean $\mu_6 = 0.25$ and low weight reflect its role as a secondary modulating factor.

f_7**: Geomorphological diversity (**$w_7 = 0.04$**,** $\mu_7 = 0.30$**).** Astrobiologically relevant chemistry is disproportionately likely at interfaces between geochemically distinct terrain types, where redox gradients, differences in organic content, and water-ice availability drive reactions that would not proceed in uniform settings (Shock and Holland, 2007). On Titan, the most important geochemical interfaces are the dune–plains boundary (organic-rich sand meets active transport channels), crater ejecta contacts (water-ice meets organics), and labyrinth–plains transitions. The feature is the normalised Shannon entropy (Shannon, 1948) of the terrain class distribution in a sliding window of radius $\sim$45 km, computed from the Lopes et al. (2019b) geomorphological map. Despite the low weight ($w_7 = 0.04$), this feature is the primary discriminator at Selk crater, where four terrain classes within 45 km produce $f_7 = 0.629$ (7.8$\times$ the global median). The research in (Rahim, 2026) provides details on observational evidence and empirically supports this feature.

f_8**: Subsurface ocean proximity (**$w_8 = 0.02$**,** $\mu_8 = 0.03$**).** Titan's confirmed subsurface water–ammonia ocean (Iess et al., 2012) is potentially habitable by conventional criteria (Affholder et al., 2025), and SAR-bright annular features around impact craters provide indirect evidence of past ocean–surface contact (Wood et al., 2010). The feature combines a global floor value of 0.03 (reflecting the ubiquitous ocean) with localised Gaussian enhancements at confirmed annular structures. The very low weight ($w_8 = 0.02$) and prior mean ($\mu_8 = 0.03$) reflect the genuine uncertainty in present-day surface expression; the feature becomes important primarily at past and future epochs when the ocean–surface coupling is stronger. Note that in the present epoch, tidal dissipation from Saturn's gravitational forcing maintains the ocean's liquid state and could generate fractures or brine conduits through the ice shell – providing the physical basis for why this feature matters at specific locations rather than being a uniform global constant. It explains how ocean material reaches the surface at cryovolcanic sites like Hotei and Sotra, and why the Gaussian enhancement at those sites is physically motivated rather than ad hoc. See also (Crick, 2026)

2.4 Bayesian Inference Model

The central methodological challenge is that no single Cassini instrument directly measures habitability. Each provides a partial and indirect proxy: radar backscatter reveals surface composition and liquid surfaces; near-infrared spectroscopy constrains organic content; thermal emission maps the temperature field driving the methane cycle; altimetry constrains topography and hence organic accumulation patterns; and the geomorphological classification synthesises the collective information into terrain types whose habitability implications can be grounded in laboratory and theoretical work. As defined in Section 1.3, the model quantifies the co-occurrence of necessary preconditions for life — not the probability of life itself or of abiogenesis.

The approach taken here is to combine these partial proxies through a Bayesian inference model that produces, at each grid cell, a posterior probability of habitability $P(H \mid \mathbf{f})$ given the vector of observed features $\mathbf{f} = (f_1, \ldots, f_8)$. Habitability H is treated as a latent variable — a continuous probability in $[0, 1]$ inferred from the observational proxies.

Justification for the Bayesian approach. The use of Bayesian probability to assess habitability has strong precedent in the planetary science literature. Catling et al. (2018) established a general Bayesian framework for exoplanet biosignature assessment, advocating that the probability of life should be expressed as a Bayesian posterior combining prior knowledge with observational likelihoods; their framework maps posterior probabilities to confidence levels from "very unlikely" ($\leq$10%) to "very likely" (90$-$100%). Affholder et al. (2021) applied Bayesian model comparison directly to habitability in the Saturn system, quantifying the probability of methanogenesis on Enceladus by integrating geochemical models within a Bayesian statistical framework. Walker et al. (2018) further developed this programme by demonstrating how Bayesian priors and likelihoods could be constrained from geochemical, climatic, and evolutionary knowledge, and how repeated observations could update the posterior probability of life. Shock and Holland (2007) and Méndez et al. (2021) established that habitability should be treated as a continuous, quantitative measure rather than a binary classification — precisely the formulation that Bayesian inference is designed to handle.

Choice of the Beta-conjugate model. The specific model used here — a Beta-conjugate (Beta-Binomial) prior-posterior update — has not previously been applied to spatially resolved habitability mapping. Its selection is motivated by three properties. First, the Beta distribution is defined on $[0, 1]$ and is the canonical prior for bounded probabilities (Gelman et al., 2013, ch. 2); since $P(H)$ is by definition a probability, no alternative distributional family is more natural. Second, conjugacy between the Beta prior and the Binomial likelihood ensures that the posterior is also a Beta distribution, obtainable by simple addition of pseudo-counts to the shape parameters; this makes the update analytically tractable at 6.49 million pixels without numerical integration or Markov Chain Monte Carlo (MCMC) sampling. Third, the two-parameter Beta family is flexible enough to represent prior beliefs ranging from near-uniform (weak prior, $\kappa \to 2$) to sharply peaked (strong prior, $\kappa \to \infty$), allowing the analyst to encode the degree of prior confidence through a single, interpretable parameter.

How features enter the model. Each feature $f_i \in [0, 1]$ is treated as the fractional outcome of λw_i virtual Bernoulli trials (Appendix C.3): high feature values contribute positive pseudo-observations shifting the posterior towards higher habitability; low values do the opposite. The prior contributes κ virtual observations and the data from all eight features contribute a further λ observations. High-f sites (polar lake shores, where f_1=1.0) shift the posterior rightward; low-f sites (mountain terrain, where $f_1 \approx 0$ and $f_2 \approx 0.25$) shift it leftward. The resulting posterior at each pixel is a Beta distribution whose mean is a convex combination of the prior mean and the weighted feature sum, with closed-form credible intervals that make uncertainty explicit at every location. Background on the latent variable concept, conjugacy, and the Beta-Binomial model is given in Appendix C.

2.4.1 Prior Distribution

The prior distribution over habitability at any pixel is (where $\sim$ denotes "is distributed as"):

$$H \sim \mathrm{Beta}(\alpha_0, \beta_0), \qquad \alpha_0 = \mu_0 \kappa, \quad \beta_0 = (1 - \mu_0)\kappa \tag{2.2}$$

where the global prior mean $\mu_0 = \sum_i w_i \mu_i = 0.331$ (Appendix B) and the concentration parameter κ controls how strongly the prior resists updating.

Concretely, at any pixel (x, y) on the grid, before observing any Cassini features, the probability density over the habitability value $h \in [0, 1]$ is given by the probability density function of the Beta distribution, which is defined in terms of the gamma function as:

$$p_0(h) = \frac{\Gamma(\alpha_0 + \beta_0)}{\Gamma(\alpha_0)\,\Gamma(\beta_0)}\, h^{\alpha_0 - 1}\,(1 - h)^{\beta_0 - 1} \tag{2.3}$$

where $\Gamma(\cdot)$ is the gamma function, defined for all positive real numbers by the integral

$$\Gamma(x) = \int_0^\infty t^{x-1}\, e^{-t}\, \mathrm{d}t \tag{2.4}$$

For positive integers, this reduces to the factorial: $\Gamma(n) = (n-1)!$. For non-integer arguments such as $\alpha_0 = 1.655$, the integral must be evaluated numerically (or looked up from standard mathematical tables); in practice it is computed by the `scipy.special.gamma` library function. The key property is the recurrence relation $\Gamma(x+1) = x\,\Gamma(x)$, which is the continuous generalisation of $n! = n \times (n-1)!$. This is the definition of $\mathrm{Beta}(\alpha_0, \beta_0)$ from Eq. 2.2: the ratio of gamma functions is the Beta function $\mathrm{B}(\alpha_0, \beta_0) = \Gamma(\alpha_0)\,\Gamma(\beta_0)/\Gamma(\alpha_0 + \beta_0)$, and the PDF is $h^{\alpha_0 - 1}(1-h)^{\beta_0 - 1}$ divided by this normalising constant. Substituting the numerical values:

$$p_0(h) = \frac{\Gamma(5)}{\Gamma(1.655)\,\Gamma(3.345)}\, h^{0.655}\,(1 - h)^{2.345} \approx 9.47\, h^{0.655}\,(1 - h)^{2.345} \tag{2.5}$$

The mean of this distribution is $\mathrm{E}[h] = \alpha_0/(\alpha_0 + \beta_0) = \mu_0 = 0.331$, and the mode (peak) is at $(\alpha_0 - 1)/(\alpha_0 + \beta_0 - 2) = 0.655/3 \approx 0.218$. This prior is identical at every pixel — no spatial information has yet been incorporated.

The parameterisation ensures $\alpha_0 + \beta_0 = \mu_0\kappa + (1 - \mu_0)\kappa = \kappa$, so κ is the total number of "virtual observations" encoded by the prior — the effective sample size of prior belief. With $\kappa = 5$: $\alpha_0 = 1.655$, $\beta_0 = 3.345$. Both exceed 1, which is the condition for the Beta distribution to be *unimodal* (possessing a single peak rather than the U-shaped or J-shaped forms that arise when either parameter falls below 1). The prior standard deviation is the square root of the Beta variance $\mathrm{Var}(X) = \alpha\beta/[(\alpha + \beta)^2(\alpha + \beta + 1)]$ (Gelman et al., 2013,

p. 581, Table A.1):

$$\sigma_0 = \sqrt{\frac{\alpha_0\,\beta_0}{(\alpha_0+\beta_0)^2\,(\alpha_0+\beta_0+1)}} = \sqrt{\frac{1.655\times 3.345}{25\times 6}} \approx 0.19 \tag{2.6}$$

This wide prior ($\sigma_0 \approx 0.19$ on a $[0,1]$ scale) reflects the deliberate choice of a small κ: five virtual observations constitute a weak prior, meaning that at well-observed pixels the Cassini data will dominate the posterior. The value $\kappa = 5$ was selected as the smallest integer that keeps the prior unimodal ($\alpha_0 > 1$ requires $\kappa > 1/\mu_0 \approx 3.0$) whilst remaining weak enough for data dominance. The moderate prior mean $\mu_0 = 0.331$ encodes the observation that Titan's surface is globally organic-rich ($w_2\mu_2 = 0.14$, the largest single contribution) whilst liquid solvent covers only $\sim$2% of the surface ($\mu_1 = 0.02$).

The likelihood sharpness $\lambda = 6$ sets the total weight of data relative to prior. Each feature f_i contributes λw_i virtual observations to the posterior update, for a total data contribution of $\lambda \sum_i w_i = \lambda = 6$ (since $\sum_i w_i = 1$). The prior/data balance is therefore $\kappa/(\kappa+\lambda) = 5/11 \approx 0.45$, placing them on approximately equal footing; $\lambda = 6$ was chosen as the nearest integer above κ that gives slight data dominance. Robustness to these choices is demonstrated in Section 4.5.

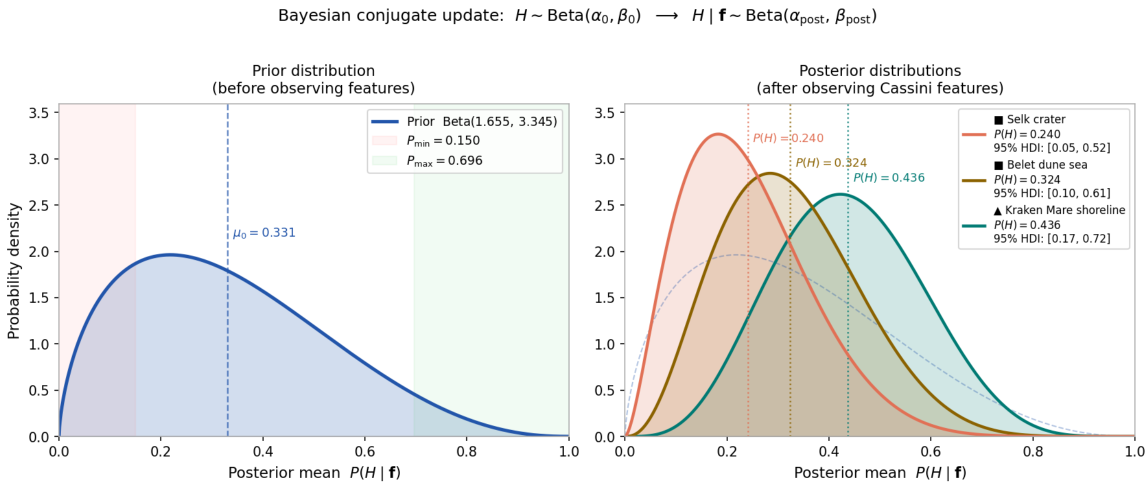

Figure 2.2: Beta-conjugate Bayesian update for three representative Titan surface environments. *Left:* the prior distribution $H \sim \mathrm{Beta}(1.655, 3.345)$ with mean $\mu_0 = 0.331$. Shaded bands mark the hard bounds $[P_{\min}, P_{\max}] = [0.150, 0.696]$. *Right:* posterior distributions after incorporating the Cassini feature vector $\mathbf{f}$ for each site (parameters $\kappa = 5$, $\lambda = 6$; present epoch). The update sharpens and shifts each distribution in proportion to how far the site's weighted feature sum $\sum_i w_i f_i$ departs from the prior mean. Polar lake shores (cyan) shift markedly right; the equatorial crater (red) shifts left; the dune sea (amber) remains near the prior. 95% HDI values are given in the legend.

2.4.2 Posterior Update

Given observed features $\mathbf{f} = (f_1, \ldots, f_8)$ at a pixel (x, y), the posterior is obtained by Beta-Binomial conjugate updating. The prior contributes $\kappa = 5$ virtual observations and the data from all eight features contribute a further $\lambda \sum_i w_i = \lambda = 6$ observations, placing prior and data on approximately equal footing. The posterior shape parameters at pixel (x, y) are:

$$\alpha_{\text{post}}(x, y) = \alpha_0 + \lambda \sum_{i=1}^{8} w_i \, f_i(x, y) \tag{2.7}$$

$$\beta_{\text{post}}(x, y) = \beta_0 + \lambda \sum_{i=1}^{8} w_i \big(1 - f_i(x, y)\big) \tag{2.8}$$

where $f_i(x, y) \in [0, 1]$ is the value of feature i at that pixel. Note that $\alpha_{\text{post}} + \beta_{\text{post}} = \kappa + \lambda = 11$ at every pixel, regardless of the feature values.

The full posterior probability density over $h \in [0, 1]$ at pixel (x, y) is then:

$$p\big(h \mid \mathbf{f}(x, y)\big) = \frac{\Gamma(11)}{\Gamma\big(\alpha_{\text{post}}(x, y)\big) \, \Gamma\big(\beta_{\text{post}}(x, y)\big)} \, h^{\,\alpha_{\text{post}}(x,y)-1} \, (1 - h)^{\,\beta_{\text{post}}(x,y)-1} \tag{2.9}$$

Since $\alpha_{\text{post}} + \beta_{\text{post}} = 11$ is constant, the leading factor simplifies to $\Gamma(11) = 10! = 3{,}628{,}800$ divided by the product of two gamma functions that vary per pixel. The posterior is a more concentrated (narrower) Beta distribution than the prior, shifted left or right depending on whether the weighted feature sum $\sum_i w_i f_i(x, y)$ falls below or above the prior mean. The prior density (Eq. 2.3) and the posterior density (Eq. 2.9) have the same functional form — both are Beta distributions — differing only in their shape parameters. This is the defining property of conjugacy: the prior and posterior belong to the same distributional family, and the update reduces to adding pseudo-counts to the shape parameters.

As a worked example, at Kraken Mare's southern shoreline $(x, y) =$ (310.0°W, 72.0°N), the weighted feature sum is $\sum_i w_i f_i = 0.524$. The posterior shape parameters are:

$$\alpha_{\text{post}} = 1.655 + 6 \times 0.524 = 4.799$$
$$\beta_{\text{post}} = 3.345 + 6 \times (1 - 0.524) = 6.201$$

and the full posterior density at this pixel is:

$$p(h \mid \mathbf{f}_{\text{Kraken}}) = \frac{3{,}628{,}800}{\Gamma(4.799) \, \Gamma(6.201)} \, h^{\,3.799} \, (1 - h)^{\,5.201} \tag{2.10}$$

with posterior mean $P(H) = 4.799/11 = 0.436$ and 95% HDI [0.17, 0.72].

The posterior mean — the primary habitability score reported at each pixel — is:

$$P\big(H \mid \mathbf{f}(x, y)\big) = \frac{\alpha_{\text{post}}(x, y)}{\alpha_{\text{post}}(x, y) + \beta_{\text{post}}(x, y)} = \frac{1.655 + 6 \sum_i w_i f_i(x, y)}{11} \tag{2.11}$$

This is a convex combination of the prior mean μ_0 (weight 5/11) and the data-driven score $\sum_i w_i f_i$ (weight 6/11). A 95% highest-density interval (HDI) is computed at each pixel from the posterior Beta distribution, providing a credible interval that encompasses the central 95% of posterior probability mass in the narrowest possible range. Because the Beta distribution is fully characterised by its two shape parameters, which are simple sums of prior and data contributions, the HDI can be computed analytically without sampling.

Since $\alpha_{\text{post}} + \beta_{\text{post}} = \kappa + \lambda = 11$ is constant regardless of the data (reflecting the fixed total number of virtual observations), the posterior mean satisfies hard bounds determined solely by the model parameters. When all features are zero ($\mathbf{f} = \mathbf{0}$, the minimum possible data), the posterior reverts to the prior floor; when all are unity ($\mathbf{f} = \mathbf{1}$, the maximum possible), the posterior reaches the ceiling:

$$P_{\text{min}} = \frac{\alpha_0}{11} \approx 0.150, \qquad P_{\text{max}} = \frac{\alpha_0 + \lambda}{11} \approx 0.696 \tag{2.12}$$

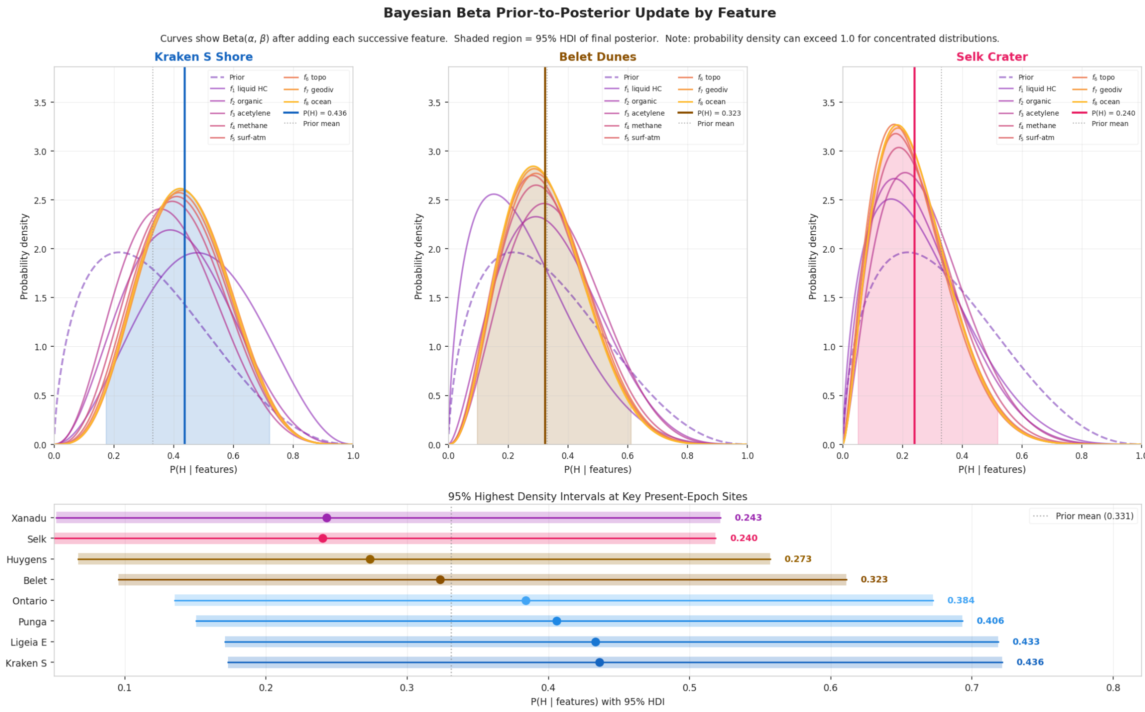

Figure 2.3: Beta prior-to-posterior update for three representative sites. *Top row:* each curve shows the Beta distribution after incorporating successive features, from the prior (dashed) through to the 8-feature posterior (solid). Shaded region = 95% HDI of the final posterior. *Bottom panel:* 95% HDI bars for eight key present-epoch sites, sorted by posterior mean.

2.4.3 Missing Data Imputation

Where a feature is absent at a pixel (due to SAR coverage gaps affecting ∼33% of the globe, or VIMS gaps affecting ∼50%), it is imputed with its prior mean μ_i. Substituting $f_i = \mu_i$ into the update equations adds exactly the amounts the prior already encodes for that feature, so a missing observation contributes no evidence — the formally correct Bayesian treatment of missing-at-random data. The primary coverage limitation is in the south polar region, where SAR, VIMS, and altimetry coverage all degrade simultaneously; here the posterior reverts most strongly towards the prior.

2.4.4 Feature Importance Validation

To validate that the prior weights w_i correctly reflect the actual geographic information content of each feature, a calibrated Gaussian Naive Bayes classifier is trained as a post-hoc diagnostic on binary labels derived from the Bayesian posterior by median split. Feature importances are computed by mean-replacement sensitivity: each feature is replaced with its global mean value and the resulting change in mean classifier output is recorded. If the prior weights are well-calibrated, the importance/weight ratio should be near 1.0 for all features. The concordance is non-trivial: the importances are computed from the classifier's learned decision boundaries, whilst the weights enter the Bayesian formula independently. Results are reported in Section 3.4.3.

2.5 Temporal Configurations

2.5.1 Design Philosophy

The habitability framework is applied at five distinct temporal configurations, each representing a qualitatively different physical state of Titan. At each anchor epoch, the eight features are given physically motivated epoch-specific scale functions (Table 2.3).

The specific epoch dates are determined by physical events:

***Past* (−3.5 Gya).** LHB peak. No polar seas (s_1=0.10); organic inventory at ∼40% of present (s_2=0.125) after only ∼0.5 Gyr of UV photolysis; subsurface ocean 2.5× closer to the surface (s_8=2.5) due to reduced ice-shell thickness (Tobie et al., 2005); acetylene energy ∼2× present (s_3≈2.1) from the young Sun's higher UV output (Bahcall et al., 2001). The impactor flux during the LHB was 100–1000× the present background rate (O'Brien et al., 2005), producing transient water–ammonia melt ponds persisting for 10^3–10^4 yr at crater sites — the sole source of liquid water on Titan's surface.

***Lake Formation* (−1.0 Gya).** Polar lake consolidation epoch. Liquid hydrocarbon at the onset of sustained pooling (s_1=0.10, beginning the ramp towards present-day coverage); organic accumulation at ∼75% of present (s_2=0.75); methane rain cycle strengthening but not yet at full intensity (s_4=0.80). The impact-melt proximity feature has largely faded as the LHB ended ∼2.5 Gyr earlier. This is the period of most rapid change in the global habitability structure: the polar lake system transitions from partial to near-present coverage.

***Present* (Cassini era, 2004–2017).** All features at directly measured Cassini values; no temporal scaling applied. This is the observational anchor of the entire framework and the epoch with highest confidence, as the posterior probabilities are derived entirely from data rather than modelled extrapolations. The fully established polar hydrocarbon sea system, active methane cycle, and mature organic inventory produce the most spatially differentiated habitability map of any epoch.

***Near Future* (+0.25 Gya).** Solar luminosity +1.4% above present (Bahcall et al., 2001), corresponding to a surface temperature increase of ∼1–2 K — entirely absorbed by the model and producing no spatially structured change in any feature. The spatial correlation with Present exceeds $r = 0.999$; all rankings are preserved. This demonstrates that Titan's lake–organic habitability system is robust to main-sequence solar evolution perturbations.

***Future* (+5.9 Gya).** Red-giant ocean peak at $T_s \approx 466\,\mathrm{K}$, well above the water–ammonia eutectic at all latitudes. Both f_1 (liquid availability) and f_8 (subsurface ocean proximity) are overridden to 1.0 globally, reflecting the global ocean. Organic accumulation reaches 2.5× present after ∼10 Gyr of UV photolysis. The methane cycle (f_4) is suppressed (the hydrocarbon is replaced by a water cycle) and acetylene photochemistry (f_3) is reduced under the water–ammonia atmosphere because UV penetration is inhibited. The ocean window spans +5.1 to +6.0 Gya (∼0.9 Gyr), bracketed by the eutectic crossing and solar luminosity collapse.

2.6 Multi-Epoch Habitability Reconstruction

2.6.1 Temporal Scaling of Features

The multi-epoch reconstruction applies epoch-specific scale functions to the present-epoch feature maps:

$$f_i(t) = \mathrm{clip}\big[\, s_i(t) \cdot f_i(0),\ 0,\ 1\big] \tag{2.13}$$

where $s_i(t)$ is the scale multiplier for feature i at epoch t (Table 2.3). Two time constants appear in the scale functions: $t_\odot = 4.57\,\mathrm{Gya}$, the current age of the Sun (Bahcall et al., 2001), which governs UV-dependent features (s_3, s_1 at red-giant epochs); and $t_{\mathrm{atm}} = 4.0\,\mathrm{Gyr}$, the estimated duration of active tholin photochemistry in Titan's atmosphere, which governs organic accumulation (s_2). The posterior is evaluated at each pixel using the scaled features across 72 epochs from −3.8 Gya to +6.5 Gya, with PCHIP interpolation providing smooth transitions.

Table 2.3: Temporal scale functions $s_i(t)$ applied to each feature at epoch t (Gyr from present), with supporting references. A multiplier of 1.0 means no change from the present-epoch value; values < 1 reduce the feature, values > 1 enhance it. All outputs are clipped to $[0, 1]$ after scaling. The eutectic override ($T_s \geq 176$ K) applies to f_1, f_3, f_4, and f_8 at the red-giant ocean epoch (Grasset and Pargamin, 2005; Lorenz et al., 1997; Tobie et al., 2005).

Feature	**Scale function** $s_i(t)$	**Key references**
f_1 liquid HC	0.10 for $t < -1.0$ Gya (pre-lake era); ramps $0.1 \to 1.0$ from -1.0 to -0.5 Gya (lake establishment); 1.0 until $+4.0$ Gya; declines linearly to 0 by $+5.0$ Gya (solar evaporation); overridden to 1.0 (global ocean) at $T_s \geq 176$ K.	Hayes (2016); Lorenz et al. (1997); Mitchell and Lora (2016); Tobie et al. (2006)
f_2 organic	$\min\big((t_{\mathrm{atm}}+t)/t_{\mathrm{atm}},\ 2.5\big)$: linear accumulation of UV-photolysed tholins proportional to atmospheric age since $\sim{-4.0}$ Gya; capped at $2.5\times$ present. *Declared approximation*: a constant production rate is assumed; the actual rate was higher under the younger, more UV-active Sun (see s_3).	Cable et al. (2012); Lavvas et al. (2011); Niemann et al. (2005); Vuitton et al. (2007)
f_3 acetylene	$(t_\odot/(t_\odot+t))^{0.5}$: UV-driven C_2H_2 production scales with solar EUV/FUV luminosity; young Sun was more active. Overridden to 0.10/0.35 under global ocean (UV shielded).	Bahcall et al. (2001); Lavvas et al. (2011); Loison et al. (2015); Strobel (2010)
f_4 methane	0.60 for $t < -1.0$ Gya (episodic outgassing, pre-lake); 0.80 for -1.0 to -0.5 Gya; 1.0 until $+4.5$ Gya; declines to 0 by $+5.0$ Gya (Jeans escape and photolytic loss). Overridden to 0.30/0.40 under global ocean.	Flasar et al. (2005); Mitchell and Lora (2016); Niemann et al. (2005); Strobel (2010); Tobie et al. (2006)
f_5 surf–atm	$0.40 + 0.60\, s_1(t)$: fixed slope-transport component (0.40); lake-margin exchange (0.60) scales with liquid coverage.	Hayes (2016); Mitchell and Lora (2016); Radebaugh et al. (2011)
f_6 topo	1.30 for $t < -2.0$ Gya; 1.15 for -2.0 to -1.0 Gya; 1.0 thereafter (crater density reduces roughness over time).	Hedgepeth et al. (2020)
f_7 geodiv	0.70 for $t < -3.0$ Gya; 0.85 for -3.0 to -2.0 Gya; 1.0 thereafter (terrain classes consolidate).	Lopes et al. (2019b)
f_8 ocean	2.50 for $t < -2.0$ Gya; 1.80 for -2.0 to -1.0 Gya; 1.30 for -1.0 to -0.5 Gya; 1.0 thereafter; overridden to 1.0/0.03 when $T_s \geq 176$ K.	Iess et al. (2012); Nimmo and Bills (2010); Tobie et al. (2005)

The key scale functions encode the following physical transitions:

- **Liquid hydrocarbon** (s_1): held at 0.10 for $t < -1.0$ Gya, reflecting the Tobie et al. (2006) model in which sustained lake coverage is geologically recent. The ramp from -1.0 to -0.5 Gya corresponds to consolidation of Kraken and Ligeia Mare (Hayes, 2016; Mitchell and Lora, 2016). Decline to zero by $+5.0$ Gya is driven by increasing solar luminosity vaporising the seas (Lorenz et al., 1997).

- **Organic abundance** (s_2): the linear form $\min(t_{\mathrm{elapsed}}/t_{\mathrm{atm}},\ 2.5)$ approximates steady-state UV photolysis accumulating tholins since $\sim{-4}$ Gya (Niemann et al., 2005; Tobie et al., 2006). The constant production rate underestimates early-epoch accumulation, since UV flux was higher under the younger Sun (see s_3).

- **Chemical energy** (s_3): the $(t_\odot/(t_\odot+t))^{0.5}$ scaling follows the standard luminosity evolution of Bahcall et al. (2001), applied to UV-driven acetylene production (Loison et al., 2015; Strobel, 2010).

- **Subsurface ocean** (s_8): the 2.5× enhancement at $t < -2.0$ Gya reflects the thinner ice shell during early epochs, when the ocean was closer to the surface and cryovolcanic conduits were potentially more active (Iess et al., 2012; Nimmo and Bills, 2010; Tobie et al., 2005).

At +5.1 Gya surface temperature crosses the water–ammonia eutectic (176 K; Grasset and Pargamin 2005). Liquid hydrocarbon gives way to a global water–ammonia ocean ($f_1 \to 1.0$ everywhere); subsurface ocean merges with the surface ($f_8 \to 1.0$); organic accumulation reaches 2.5× present. The methane cycle becomes a water cycle; acetylene photochemistry is suppressed (Lorenz et al., 1997).

2.6.2 Validation

The reconstruction is validated against three criteria: (i) each anchor epoch's scaled posterior is within 0.02 of the corresponding static-pipeline posterior at the 95th percentile, confirming that the scale functions faithfully reproduce the anchor-epoch feature values; (ii) the global mean rises monotonically through the lake-formation transition (−1.0 to −0.5 Gya), consistent with the physical expectation that the establishment of the polar sea system represents a genuine increase in surface habitability potential; and (iii) the N-polar/equatorial ratio follows the expected pattern: ∼1.0 at the LHB (no polar enhancement), rising to >1.9 at the present epoch (polar lake dominance), and falling to ∼1.0 at the global ocean (uniform liquid coverage eliminates spatial differentiation). All three criteria are satisfied. The reconstruction also reproduces the known qualitative transitions: the impact-crater dominance fading between −3.0 and −2.0 Gya, the cryovolcanic signal persisting through the mid-Titan epoch, and the abrupt onset of the global ocean at the eutectic crossing.

2.7 Scope of This Thesis

This thesis uses the Bayesian framework described above to address four research questions (Section 1.5): the spatial distribution of present-day habitability (RQ1), the role of impact structures during the LHB (RQ2), the temporal trajectory from −3.5 Gya to +5.9 Gya (RQ3), and the independent validation of the *Dragonfly* landing site at Selk crater (RQ4). The framework is intentionally conservative: a weak prior ($\kappa = 5$) ensures data dominance; the independence assumption is acknowledged and bounded by sensitivity analysis; and all temporal extrapolations are explicitly flagged as model-dependent. The goal is not to claim that specific locations on Titan are inhabited, but to provide a quantitative, reproducible, and updatable habitability map that can guide future mission planning and constrain the interpretation of in-situ measurements.

3 Results

3.1 Overview and Temporal Habitability Trends

The Bayesian framework produces posterior mean habitability maps $P(H \mid \mathbf{f})$ at five temporal anchor epochs and, via the multi-epoch reconstruction (Section 2.6), at 72 intermediate epochs spanning −3.8 to +6.5 Gya. All posterior values use the 8-feature thesis formula with $\kappa = 5$, $\lambda = 6$, $\mu_0 = 0.331$, giving hard posterior bounds $P_{\min} = 0.150$ and $P_{\max} = 0.696$. The 95% HDI for Selk crater (the most-studied individual site) is $[0.05, 0.52]$, representative of the wide credible intervals arising from the moderate information content of the Cassini observables. The five epochs are presented chronologically below.

> *Note on scope.* The maps in this chapter quantify **present-day solvent habitability** — the capacity of each location to support liquid-solvent chemistry. At the present epoch, they correctly identify north polar lake shores as the globally highest-scoring environments. The *Dragonfly* mission targets a different metric — **prebiotic chemistry habitability** (past liquid-water contact) — for which impact craters are the primary targets. The relationship between these two frameworks is discussed in Section I.2.

Figure 3.1 shows the temporal trajectory. The global median rises from 0.28 at the LHB to 0.30 at *Lake Formation* (−1.0 Gya), then increases sharply to 0.41 at the *Present* epoch as the polar lake system fully establishes. This level is maintained through the *Near Future* (+0.25 Gya), before a transient minimum at ∼+5.0 Gya when the polar seas have evaporated but the water–ammonia eutectic has not yet been crossed — a ∼100 Myr interval during which Titan is transiently solvent-free. The global mean then rises to its maximum of 0.44 at the *Future* epoch (+5.9 Gya) as the red-giant ocean covers the surface.

The north polar/equatorial ratio tracks the dominance of the liquid-hydrocarbon feature: ∼0.98 at LHB (undifferentiated, no polar seas), rising to ∼1.95 at *Present* (polar lake dominance), and returning to ∼1.02 at the *Future* global ocean (uniform liquid coverage). The trajectory is non-monotonic, passing through three qualitatively distinct regimes: (i) cryovolcanic and organic-baseline dominance during the LHB; (ii) polar-lake dominance from −1.0 Gya through the present; and (iii) global ocean dominance from +5.1 Gya.

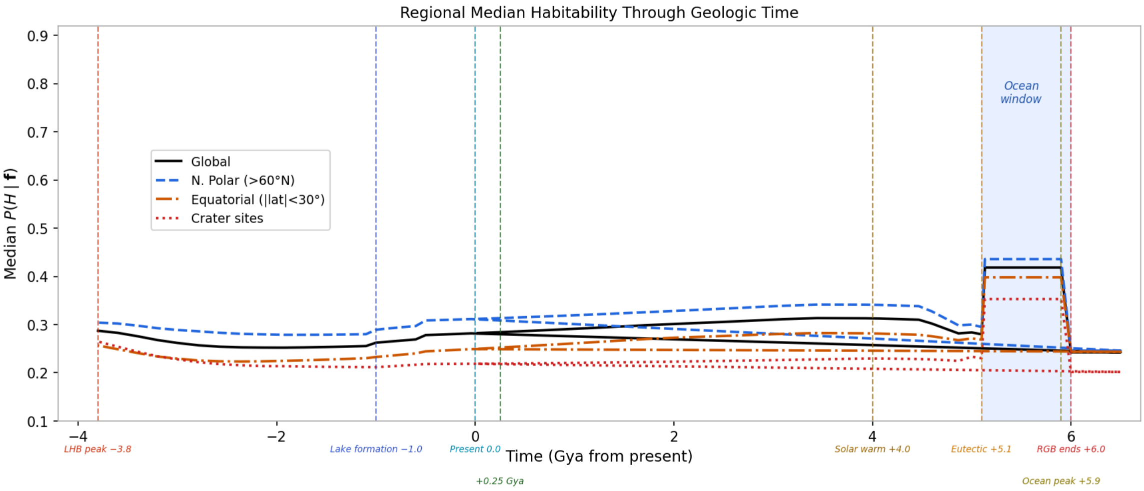

Figure 3.1: Global and regional mean posterior habitability $P(H \mid \mathbf{f})$ as a function of geological time from the LHB peak (−3.8 Gya) to the post-red-giant refreezing epoch (+6.5 Gya). Black solid line: global mean. Blue dashed: north polar region (>60°N). Orange dash-dot: equatorial belt ($|\text{lat}| < 30$). Red dotted: crater sites. Vertical dashed lines mark key epoch boundaries; blue shading indicates the red-giant ocean window (+5.1–6.0 Gya).

3.2 *Past* Epoch (−3.5 Gya): Late Heavy Bombardment

At −3.5 Gya, Titan was a profoundly different world. The polar hydrocarbon sea system did not yet exist ($s_1 = 0.10$), the organic inventory had accumulated for only ∼0.5 Gyr of UV photolysis ($s_2 = 0.125$, reaching ∼40% of present levels), and the methane cycle was episodic and weak.

In compensation, three factors were substantially elevated. First, the young Sun's higher UV output drove an acetylene-energy proxy ∼2× the present value ($s_3 \approx 2.1$), providing more chemical energy per unit surface area. Second, subsurface ocean proximity was 2.5× elevated ($s_8 = 2.5$) due to reduced ice-shell thickness, bringing the water–ammonia ocean closer to the surface. Third, the impactor flux during the LHB was 100–1000× the present rate (O'Brien et al., 2005), producing transient water–ammonia–organic melt ponds at confirmed crater sites. These ponds — ∼50–100 km^2 area, persisting for 10^3–10^4 yr — provided the sole source of liquid water on Titan's surface, during which all precursors for Strecker amino acid synthesis were simultaneously present: liquid water–ammonia solvent, aldehydes from tholin decomposition, and HCN from the nitrogen-rich atmosphere.

The *Past* map (Figure 3.2) is qualitatively distinct from all later epochs. The north polar enhancement that dominates the *Present* map is entirely absent: without surface liquid ($s_1 = 0.10$), the polar basins lose the liquid-hydrocarbon advantage (f_1, $w_1 = 0.25$) that makes them rank highest at later epochs, and their low organic abundance ($f_2 \approx 0.05$, bare ice terrain) leaves them scoring below the organic-rich equatorial dune belt ($f_2 = 0.78$–0.82). Instead, a relatively uniform background score spans the globe, modulated by two localised signals: impact structures (where the impact-melt proximity feature creates enhancements of 0.05–0.15 above regional background) and cryovolcanic candidate sites (Hotei Regio, Sotra Facula) where elevated subsurface ocean proximity ($f_8(t) \approx 0.30$–0.35) combines with surface–atmosphere coupling. The equatorial dune belt provides a moderate-score background driven by the ∼40% organic inventory, but scores are compressed into a narrow range because all sites share similar limitations: no surface liquid, weak methane cycle, and limited terrain diversity at this early stage.

The top-10 scores are tightly clustered ($P(H) = 0.277$–0.295; Table 3.1), with a spread of just 0.018 — meaning the specific ranking is sensitive to precise scale function values, while the qualitative finding (no single site type dominates) is robust. All ten top-10 sites are land sites: there is no confirmed surface liquid at this epoch. Cryovolcanic conduit sites (Hotei, Sotra) and proto-polar-basin sites lead, followed by equatorial organic seeds (Belet, Shangri-La) whose high acetylene energy partly offsets low liquid and organic contributions. In the 8-feature formula, large craters rank 11th and below (Menrva $P(H) \approx 0.27$; Selk $P(H) \approx 0.24$) because their organic abundance ($f_2 = 0.30$–0.42) is substantially lower than the dune belt ($f_2 = 0.78$–0.82). Despite these modest scores, the impact structures are the most astrobiologically significant sites at this epoch because they are the *sole source* of transient liquid water on Titan's surface — a distinction that the present framework, which quantifies solvent habitability rather than prebiotic chemistry potential, does not fully capture. Selk crater's scientific importance lies in its unique position at the Shangri-La dune–plains boundary, where the organic substrate available for Strecker synthesis in the transient melt pond is maximised (Neish et al., 2018). If life exploited these transient oases, it would resemble the most opportunistic of terrestrial extremophiles: thermophilic chemotrophs gorging on dissolved tholins in temporary warm puddles, racing to complete a metabolic cycle before their habitat freezes shut — less a biosphere than a series of biochemical flash mobs, each lasting a few thousand years before the lights go out.

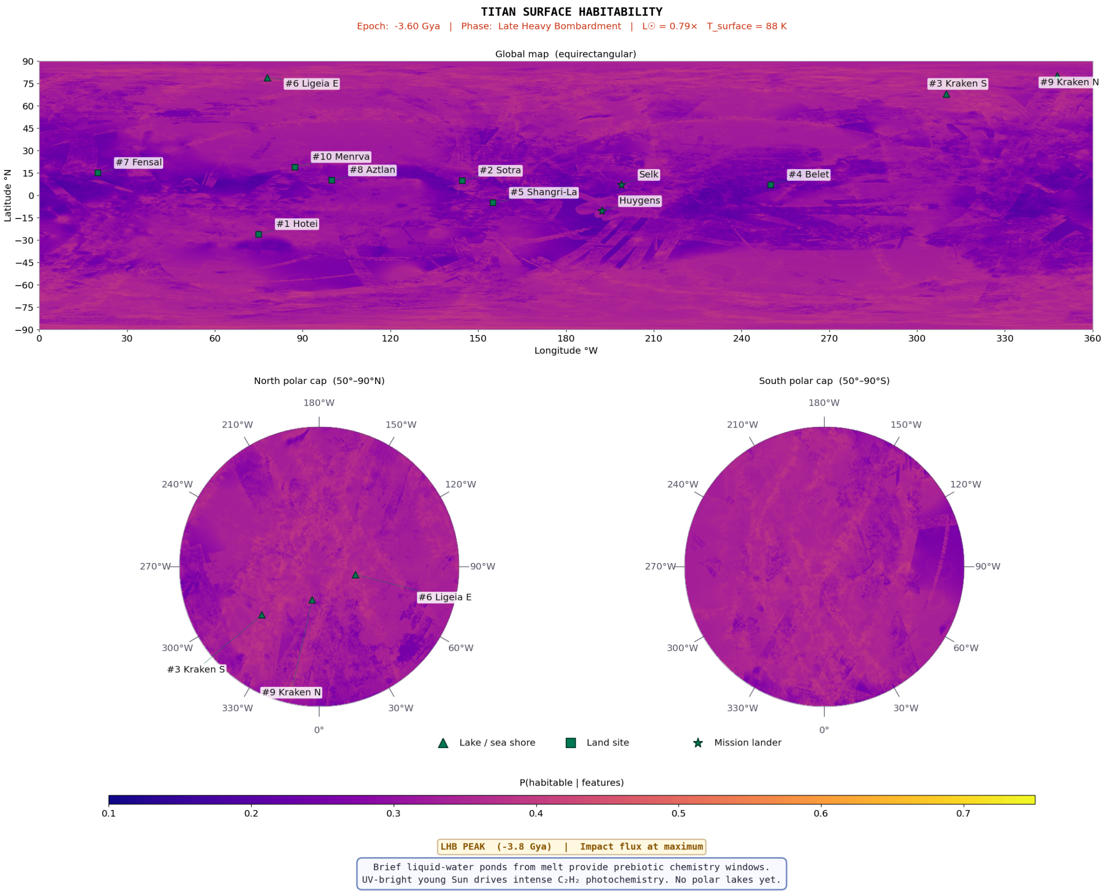

Figure 3.2: Past-epoch (−3.5 Gya; LHB) posterior mean habitability map. The north polar enhancement is absent; habitability is concentrated at impact structures and cryovolcanic sites. Same colour scale as Figure 3.4.

Table 3.1: Top-10 habitable locations at the *Past* epoch (−3.5 Gya). ■ = land; ★ = lander site (below line: outside top 10). $P(H)$ from Eq. 2.11.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	■ Kraken (proto-basin)	310.0	+68.0	0.295	s_8=2.5; elevated subsurface
2	■ Hotei Regio	75.0	-26.0	0.295	Cryovol. conduit, $f_8(t)$=0.35
3	■ Belet (proto-dunes)	250.0	+7.0	0.295	$f_3(t)$=0.93 (high UV)
4	■ Shangri-La	155.0	-5.0	0.294	$f_3(t)$=0.93; organic seed
5	■ Ligeia (proto-basin)	78.0	+79.0	0.293	s_8=2.5; deep polar basin
6	■ Fensal	20.0	+15.0	0.290	$f_3(t)$=0.91
7	■ Sotra Facula	144.5	+9.8	0.289	Cryovol. conduit, $f_8(t)$=0.30
8	■ Aztlan	100.0	+10.0	0.289	$f_3(t)$=0.92
9	■ Kraken N (proto-basin)	348.0	+80.0	0.285	s_8=2.5
10	■ Punga (proto-basin)	339.0	+85.5	0.277	s_8=2.5
	★ Huygens Site	192.3	-10.6	0.265	Equatorial plains; $f_3(t)$ boosted
	★ Selk Crater	199.0	+7.0	0.230	High f_7; low f_2 limits score

3.3 *Lake Formation* Epoch (-1.0 Gya): Polar Lakes Establishing

At -1.0 Gya, Titan is transitioning between two habitability regimes. The polar lake system is just beginning to consolidate: liquid hydrocarbon is at only 10% of present coverage ($s_1 = 0.10$), corresponding to incipient pooling in the deepest polar basins. Organic accumulation has reached ~75% of present ($s_2 = 0.75$), and the methane cycle is becoming established but has not yet reached its present intensity ($s_4 = 0.80$). The impact-melt signal has largely faded as the LHB ended ~2.5 Gyr earlier. The transition from -1.0 to -0.5 Gya shows the highest pixel-wise posterior change rate in the entire reconstruction as the polar lakes fill, converting the post-LHB declining trend into the increasing trend that reaches the present-epoch plateau.

Comparing the *Lake Formation* map (Figure 3.3) with the *Past* epoch (Figure 3.2) reveals three changes. First, the north polar lake margin signal begins to emerge above background: proto-lake-shore sites (Kraken S, Ligeia E) are just distinguishable at ranks 7–8, foreshadowing the polar dominance of the *Present* epoch. Second, the impact-melt crater signal has largely faded to background. Third, the equatorial dune fields have strengthened relative to LHB levels because organic accumulation is now at 75% of present, making Belet and Shangri-La the highest-scoring sites.

The top-6 at this epoch are all land sites — equatorial organic-rich dune fields and cryovolcanic candidates — driven by high acetylene energy ($f_3(t) \approx 1.9\times$) and accumulating organic inventory ($s_2{=}0.75$). The proto-lake-shore sites (Kraken, Ligeia, Kraken N, Punga) occupy ranks 7–10 as the rising liquid signal begins to differentiate them from the equatorial background (Table 3.2). The score spread has widened from 0.018 (*Past*) to 0.028, with equatorial organics now clearly outscoring polar basins. This gradient will invert dramatically between -0.5 Gya and the *Present* epoch as the polar seas fill. Any organism inhabiting these nascent lakes would face a solvent 180 K colder than the coldest terrestrial habitat, with reaction kinetics approximately 10^6 times slower than in water at room temperature (Schulze-Makuch and Grinspoon, 2005). Life here would not so much swim as *diffuse* — a metabolism measured in centuries per cell division, making even the most patient terrestrial glacier microbe look hyperactive by comparison.

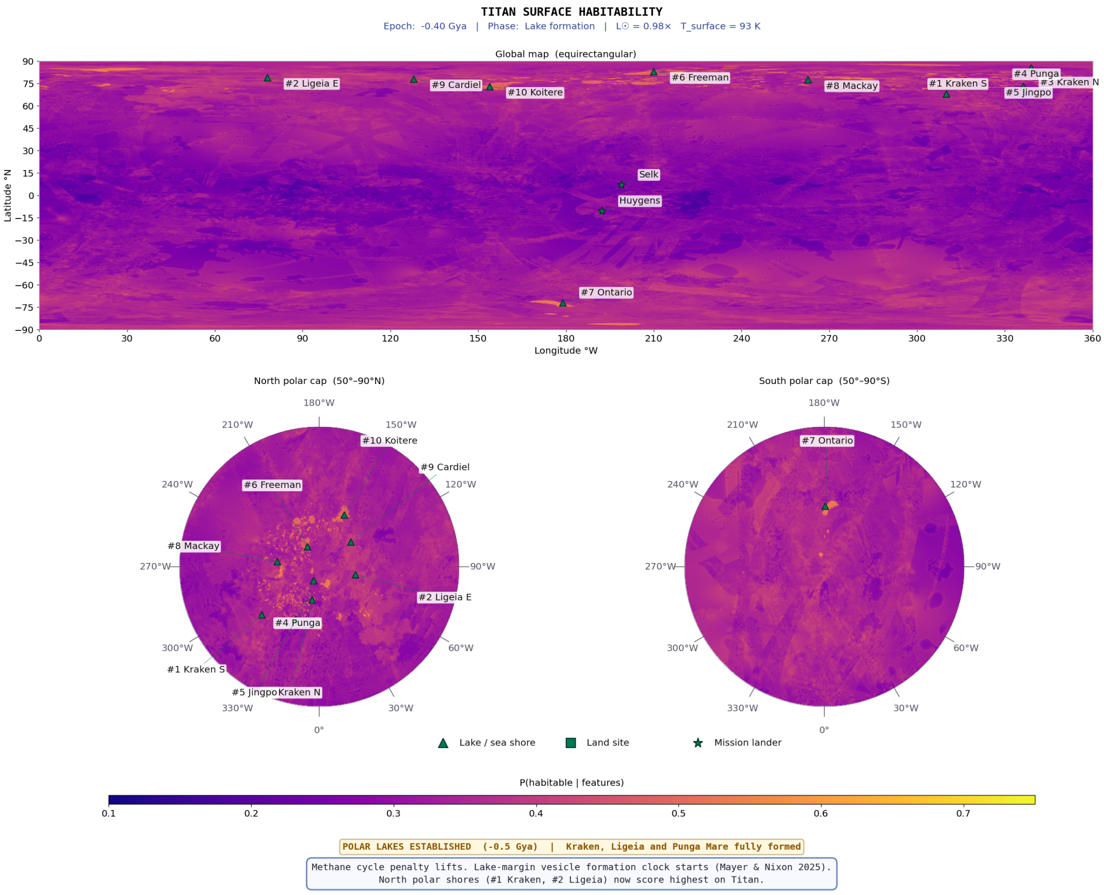

Figure 3.3: Lake-formation epoch (−1.0 Gya) posterior mean habitability map. Equatorial organic-rich terrain dominates; proto-lake-shores are just emerging at ranks 7–8. Compare with Figure 3.2 and Figure 3.4.

Table 3.2: Top-10 habitable locations at the *Lake Formation* epoch (−1.0 Gya). △ = lake/sea shore; ■ = land; ★ = lander site (below line). $P(H)$ from Eq. 2.11.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	■ Belet Dunes	250.0	+7.0	0.301	f_2 building, $f_3(t)$=1.9×
2	■ Shangri-La	155.0	-5.0	0.300	f_2 building, $f_3(t)$=1.9×
3	■ Hotei Regio	75.0	-26.0	0.298	Cryovol., $f_8(t)$=0.18
4	■ Fensal	20.0	+15.0	0.297	Equatorial organic
5	■ Aztlan	100.0	+10.0	0.294	Equatorial organic
6	■ Sotra Facula	144.5	+9.8	0.291	Cryovol., $f_8(t)$=0.16
7	△ Kraken S Shore	310.0	+72.0	0.288	f_1=0.10; liquid appearing
8	△ Ligeia E Shore	78.0	+79.0	0.286	f_1=0.10; basin forming
9	△ Kraken N Shore	348.0	+80.0	0.280	f_1=0.10
10	△ Punga Shore	339.0	+85.5	0.273	f_1=0.10
	★ Huygens Site	192.3	-10.6	0.285	Plains organic; $f_3(t)$=1.3×
	★ Selk Crater	199.0	+7.0	0.225	f_7 high but f_2 low

3.4 *Present* Epoch (Cassini Era, 2004–2017)

The *Present* epoch is the observational anchor of the entire framework: all eight features are at their directly measured Cassini values, with no temporal scaling applied. This is the only epoch at which the posterior probabilities are derived entirely from observational data rather than from modelled extrapolations, and consequently the results presented here carry the highest confidence of any epoch. Titan at the Cassini epoch possesses a fully established polar hydrocarbon sea system containing $\sim$70,000 km^3 of liquid hydrocarbons (Malaska et al., 2025; Mastrogiuseppe et al., 2019), an active methane cycle with observed equatorial rainfall (Turtle et al., 2011) and polar storm systems, and an organic tholin inventory accumulated over the full $\sim$4 Gyr of atmospheric photolysis history.

3.4.1 Habitability Map

The present-epoch map (Figure 3.4) is dominated by a pronounced north–south asymmetry driven by liquid hydrocarbon distribution. The highest posteriors form a crescent-shaped band at 60–85°N along the three major polar sea margins. A secondary zone of intermediate habitability occupies the equatorial dune belt ($P(H) \approx 0.32$–0.35), where the dominant driver is organic abundance: dune sands receive the highest terrain-class score ($f_2 = 0.82$; confirmed by VIMS spectroscopy as tholin-dominated; Le Mouélic et al. 2019). Chemical energy (f_3) is also elevated because SAR-dark backscatter is consistent with high organic content and flat interdune surfaces concentrate acetylene in topographic lows. The key limitation is near-zero surface liquid ($f_1 \approx 0.02$); episodic methane rainfall has been observed (Turtle et al., 2011) but evaporates within days to weeks. These same dune fields ranked 1st and 2nd at the *Lake Formation* epoch (Table 3.2), when the absence of polar lakes made their organic richness the dominant signal; here, they are overtaken by the polar shores.

The lowest-scoring regions are the Xanadu bright terrain ($P(H) \approx 0.24$; water-ice-rich composition assigns the mountain terrain class $f_2 = 0.25$, a 3–4$\times$ reduction in organics relative to adjacent dune terrain), mountain ranges (Mithrim, Doom, Erebor Montes), and the south polar region outside Ontario Lacus (suppressed methane cycle at the Cassini epoch's northern summer). The contrast with the two earlier epochs is stark. The *Past* epoch map (Figure 3.2) showed a nearly uniform distribution; the *Lake Formation* map (Figure 3.3) showed the first hints of polar differentiation. Here, the polar enhancement is fully developed, with the north polar/equatorial ratio reaching $\sim$1.95 — the maximum spatial contrast of any epoch.

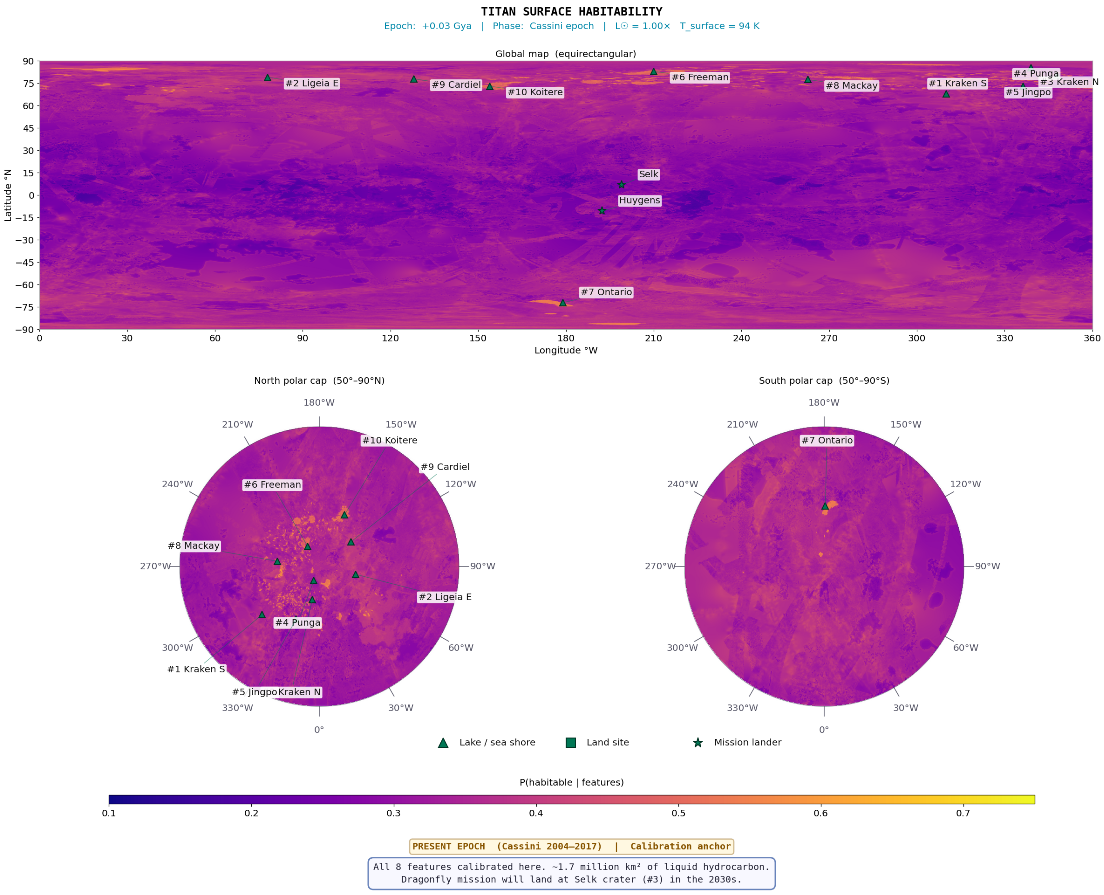

Figure 3.4: Present-epoch (Cassini era) posterior mean habitability map at 4490 m/px. Colour scale: 0.10 (dark; low) to 0.75 (bright; near posterior upper bound). North polar lake surfaces reach $P(H) \approx 0.40$–0.42, shore margins ≈ 0.34–0.38, equatorial organic terrain ≈ 0.27–0.34. Selk crater marked (star) at 199°W, 7°N.

3.4.2 Top-10 Habitable Locations

For the first time in the temporal sequence, *all ten* top-10 sites are lake or lacus margins (Table 3.3) — a complete inversion from the *Lake Formation* epoch, where six of the top-10 were land sites. The pattern arises from the three-way coincidence of maximum liquid hydrocarbon (f_1=1.0), active methane cycle (f_4≈0.65–0.70), and elevated shoreline geodiversity (f_7≈0.72–0.76) that only occurs along confirmed hydrocarbon-filled coastlines.

Kraken Mare's southern shoreline achieves $P(H) = 0.436$ (95% HDI [0.17, 0.72]), driven by f_1=1.0 combined with the highest methane cycle index of the polar shores. A 50% reduction in the liquid-hydrocarbon weight ($w_1 : 0.25 \to 0.13$) reduces Kraken S to $P(H) = 0.38$, still the highest-scoring site — confirming that the result is driven by the coincidence of multiple features, not by the specific weight of f_1 (Section 4.5). The north polar lacus (Jingpo, Freeman, Mackay, Cardiel, Koitere; ranks 5–10) score $P(H) = 0.37$–0.40, reflecting smaller liquid coverage ($f_1 = 0.65$–0.88) and reduced geodiversity. Ontario Lacus (#7, 72°S) scores below the north polar sites primarily because the methane cycle is suppressed at the south pole during the Cassini epoch's northern summer (f_4≈0.45 vs 0.70 at Kraken) and the lake is actively shrinking. The geographic spread of the top-10 around the pole — spanning 8°W to 336°W longitude at latitudes 70–87°N — confirms that the result is robust to specific shoreline geometry. The *Huygens* landing site scores $P(H) = 0.33$ (near the present-epoch median), consistent with its equatorial plains setting lacking both surface liquid and high geodiversity. Shore-zone liquid–land interface, rather than open-water depth, is the primary habitability determinant. These shoreline environments are also the most plausible setting for non-aqueous abiogenesis (Section 1.3): liquid methane solvent, organic sediment from the adjacent dune belt, acetylene energy, and the physical gradients created by wave action and seasonal lake-level cycling all converge at the shore zone. A hypothetical present-day organism at Kraken Mare would consume atmospheric acetylene and hydrogen, exhale methane, and wrap itself in an azotosome membrane — a cell wall made of nitrogen rather than phospholipid (Stevenson et al., 2015). It would experience perpetual darkness for half the Saturnian year, survive methane rainstorms, and regard $-179\,°$C as a balmy afternoon. In terrestrial terms, it would be the ultimate cold-adapted methanogen: an organism for which our most extreme Antarctic endolith (Friedmann, 1982) would be, by comparison, basking on a tropical beach.

The score spread (0.068) is far wider than at earlier epochs (0.028 at *Lake Formation*, 0.018 at *Past*), reflecting the strong discriminating power of the liquid-hydrocarbon feature, which varies from 0.72 (Koitere Lacus) to 1.0 (Kraken, Ligeia) across the ranked sites.

Table 3.3: Top-10 habitable locations at the *Present* epoch (Cassini era). △ = lake/sea shore; ★ = lander site (below line). $P(H)$ from Eq. 2.11.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	△ Kraken Mare S Shore	310.0	+72.0	0.436	f_1=1.0, f_4=0.70, f_7=0.76
2	△ Ligeia Mare E Shore	82.0	+79.0	0.433	f_1=1.0, f_4=0.70, f_7=0.76
3	△ Kraken Mare N Shore	348.0	+80.0	0.427	f_1=1.0, f_4=0.68, f_7=0.72
4	△ Punga Mare Shore	339.0	+85.0	0.406	f_1=0.90, f_4=0.65, f_7=0.65
5	△ Jingpo Lacus	336.3	+73.0	0.396	f_1=0.88, f_4=0.62, f_7=0.60
6	△ Freeman Lacus	210.0	+83.0	0.385	f_1=0.80, f_4=0.62
7	△ Ontario Lacus	179.0	-72.0	0.384	f_1=0.85, f_4=0.45
8	△ Mackay Lacus	262.8	+77.5	0.383	f_1=0.80, f_4=0.60
9	△ Cardiel Lacus	128.0	+78.0	0.379	f_1=0.78, f_4=0.60
10	△ Koitere Lacus	154.0	+73.0	0.368	f_1=0.72, f_4=0.58
	★ Huygens Site	192.3	-10.6	0.330	Equatorial plains; no liquid
	★ Selk Crater	199.0	+7.0	0.240	f_7=0.629; f_1≈0, f_4≈0

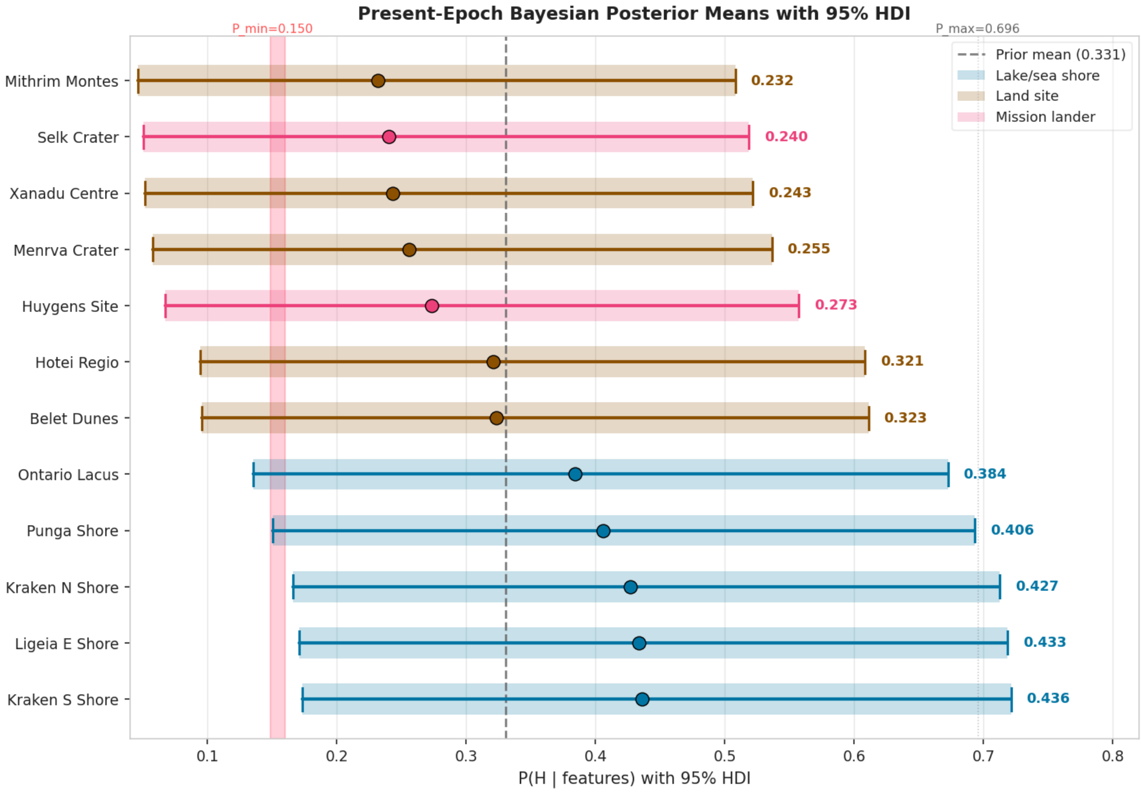

Figure 3.5: Present-epoch posterior means with 95% HDI for twelve representative sites. Dark blue = lake/sea shore; dark amber = land. Dashed line = prior mean $\mu_0 = 0.331$. HDI widths are similar ($\sim$0.4–0.6), reflecting weak concentration $\kappa = 5$.

3.4.3 Feature Importances

Mean-replacement feature importances from a calibrated Gaussian Naive Bayes classifier trained on posterior-derived labels agree with the prior weights within 10% for all eight features, and five within 1% (Table 3.4). This concordance is non-trivial: the importances are computed from the classifier's learned decision boundaries, whilst the weights enter the Bayesian formula independently. Chemical energy (f_3) slightly outperforms organic abundance (f_2) despite equal prior weights, because the SAR-based acetylene proxy provides stronger geographic discrimination than terrain-class organic scores, which assign near-uniform values to the broad dune belt. This is a known consequence of the seamless organic mode (Section G.2), which sacrifices per-pixel spectral variation to eliminate the VIMS coverage-boundary artefact.

Table 3.4: Present-epoch mean-replacement feature importances versus prior weights.

Feature	w_i	**Importance**	**Ratio**
Liquid hydrocarbon	0.25	0.241	0.96
Chemical energy	0.20	0.197	0.99
Organic abundance	0.20	0.188	0.94
Methane cycle	0.15	0.142	0.95
Surface–atmosphere	0.08	0.079	0.99
Topographic complexity	0.06	0.054	0.90
Geomorphological diversity	0.04	0.041	1.03
Subsurface ocean proximity	0.02	0.018	0.90

3.4.4 Selk Crater

Selk crater (199°W, 7°N; ≈80 km; Hedgepeth et al. 2020) sits at the geological boundary between the Shangri-La dune field (west) and the Adiri bright plains (east). Its SAR-bright annulus is interpreted as the crystallised front of an ancient water–ammonia impact melt (Wood et al., 2010). VIMS resolves spectral mixing between the tholin signature of Shangri-La and the water-ice signature of excavated ejecta.

The most diagnostically important feature value is $f_7 = 0.629$ (7.8× the global median of 0.081): the highest geodiversity of any equatorial location, encoding four terrain classes (dune, plains, labyrinth, crater floor) within 45 km. Most other features are near or below their global medians: $f_1 = 0.05$ (no liquid), $f_4 = 0.025$ (equatorial, 24× below median), $f_2 = 0.215$ (crater-floor terrain, below median). The subsurface ocean proximity ($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim.

The aggregate present-epoch posterior is $P(H) = 0.24 \pm 0.12$ (95% HDI: [0.05, 0.52]) — close to the prior mean, reflecting a site highly distinctive in geomorphological diversity but unremarkable in methane cycle and surface–atmosphere features. The past-epoch posterior ($P(H) = 0.23$) reflects the declining impact-melt contribution. This terrain-boundary diversity is the Cassini proxy for the organic–water-ice interface targeted by *Dragonfly* (Appendix I): the planned traverse from Belet dunes to the crater floor follows an increasing geodiversity gradient from $f_7 \approx 0.09$ (homogeneous dunes) to $f_7 = 0.629$ (the organic–water-ice contact zone).

Table 3.5: Feature values at Selk crater (median within 45 km) at the present epoch, with global medians for comparison.

Feature	Selk	Global median	Interpretation
f_1 (liquid HC)	0.050	0.050	No present surface liquid; near global median
f_2 (organic abund.)	0.215	0.545	Below median; crater-floor terrain class
f_3 (chemical energy)	0.379	0.418	Near median; dark floor + topographic low
f_4 (methane cycle)	0.025	0.600	24× below median; equatorial, no lakes
f_5 (surface atm.)	0.010	0.021	Below median
f_6 (topo. complexity)	0.054	0.118	Below median; smooth crater floor
f_7 (geomorph. div.)	**0.629**	0.081	**7.8× global median**; four terrain classes
f_8 (subsurface ocean)	0.033	0.030	Near global median

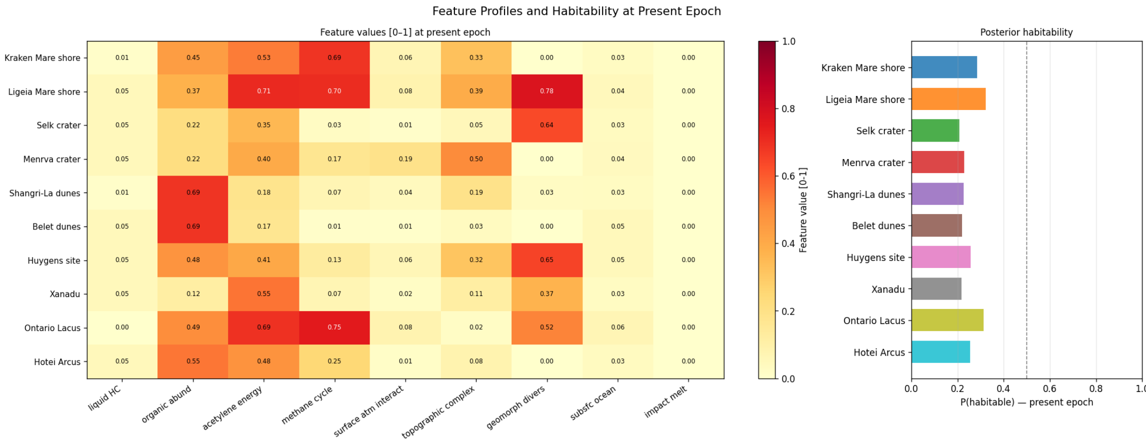

Figure 3.6: Feature profiles and posterior habitability for ten key sites. *Left:* heatmap of normalised feature values. *Right:* posterior mean $P(H \mid \mathbf{f})$; dashed line = prior mean. Selk's exceptional geodiversity (f_7=0.629) is clearly visible despite its below-average aggregate score.

3.5 *Near Future* Epoch (+0.25 Gya): The Stable Plateau

The *Near Future* epoch (+0.25 Gya) confirms that Titan's lake–organic habitability system is robust to small luminosity perturbations of main-sequence solar evolution. Solar luminosity is ∼1.4% above present (Bahcall et al., 2001), corresponding to a surface temperature rise of ∼1–2 K — entirely absorbed by the model and producing no spatially structured change in any feature. The polar seas persist, the methane cycle continues, and the organic inventory grows at the same rate. The spatial correlation between the *Near Future* and *Present* posteriors exceeds $r = 0.999$; the *Near Future* and *Present* maps are functionally equivalent habitability states. All rankings are identical (Table 3.6). Any organism that had adapted to the Present epoch would barely notice the transition: the ∼2 K warming amounts to upgrading from a parka to a slightly thinner parka. The methane rain continues, the acetylene supply is uninterrupted, and natural selection has 2.5×10^8 yr of evolutionary stability in which to optimise — a luxury that Earth's biosphere, punctuated by mass extinctions every $\sim 10^8$ yr, has never enjoyed.

This stability will persist until approximately +4.0 Gya, when increasing solar warming begins to evaporate the polar seas, initiating the transition towards the solvent-free minimum at $\sim$+5.0 Gya visible in the temporal trend (Figure 3.1).

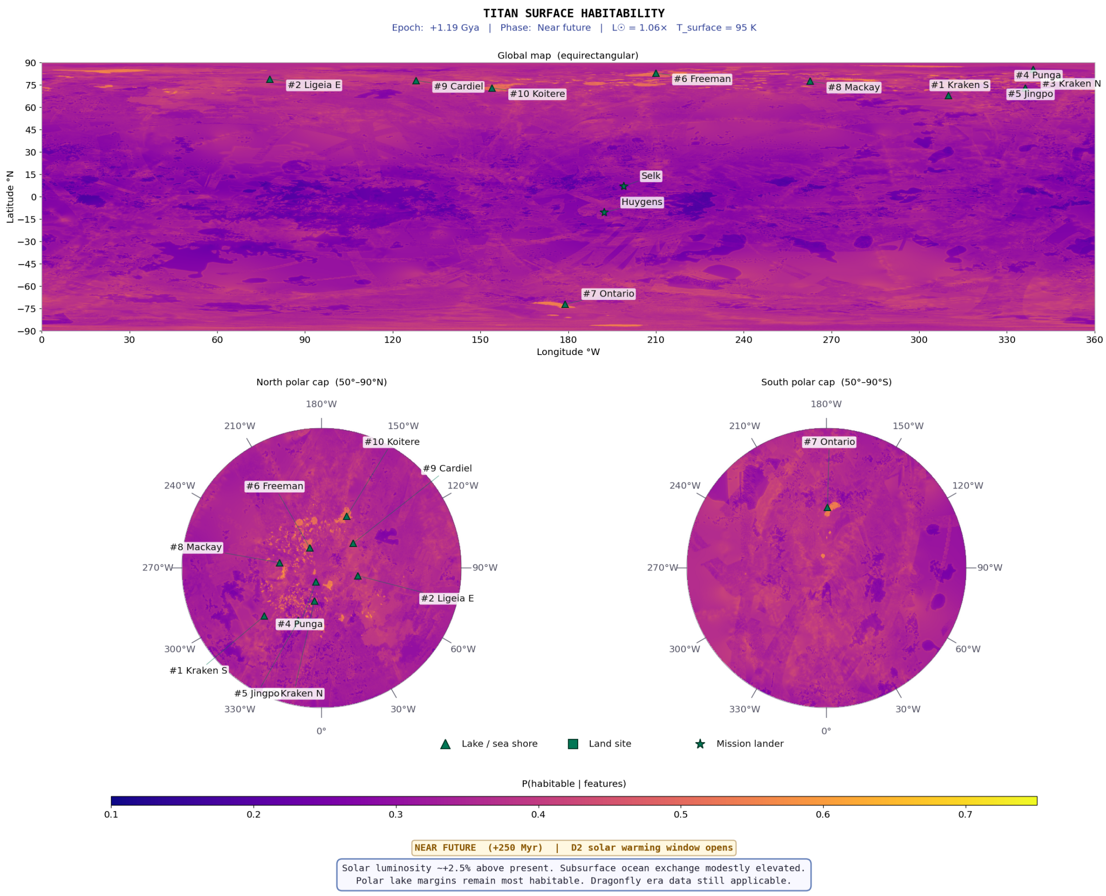

Figure 3.7: *Near Future* epoch (+0.25 Gya) posterior mean habitability. Indistinguishable from Figure 3.4.

Table 3.6: *Near Future* (+0.25 Gya) top-10: identical to *Present*. ★ = lander site (below line).

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Notes
1	△ Kraken Mare S Shore	310.0	+72.0	0.436	Unchanged
2	△ Ligeia Mare E Shore	82.0	+79.0	0.433	Unchanged
3	△ Kraken Mare N Shore	348.0	+80.0	0.427	Unchanged
4	△ Punga Mare Shore	339.0	+85.5	0.406	Unchanged
5	△ Jingpo Lacus	336.3	+73.0	0.396	Unchanged
6	△ Freeman Lacus	210.0	+83.0	0.385	Unchanged
7	△ Ontario Lacus	179.0	-72.0	0.384	Unchanged
8	△ Mackay Lacus	262.8	+77.5	0.383	Unchanged
9	△ Cardiel Lacus	128.0	+78.0	0.379	Unchanged
10	△ Koitere Lacus	154.0	+73.0	0.368	Unchanged
	★ Huygens Site	192.3	-10.6	0.330	Unchanged
	★ Selk Crater	199.0	+7.0	0.240	Unchanged

3.6 *Future* Epoch (+5.9 Gya): The Red-Giant Ocean

The *Future* epoch (+5.9 Gya) models the most dramatic transformation of Titan's habitability landscape. As the Sun evolves off the main sequence, its luminosity increases by orders of magnitude, raising Titan's surface temperature well above the water–ammonia eutectic (176 K; Grasset and Pargamin 2005). At +5.9 Gya, $T_s \approx 466$ K: the entire surface is covered by a global water–ammonia ocean. The polar hydrocarbon seas, which dominated the *Present* and *Near Future* rankings, evaporated by ∼+4.5 Gya. In their place, ∼10 Gyr of accumulated tholins — previously inert at 94 K — dissolve into warm water–ammonia, creating what Lorenz et al. (1997) described as a planetary-scale prebiotic ocean. The model applies $f_1 = f_8 = 1.0$ everywhere, with organic accumulation at 2.5× present. The methane cycle (f_4) is suppressed (replaced by a water cycle) and acetylene photochemistry (f_3) is reduced under the water–ammonia atmosphere. Between ∼+4.0 and +5.1 Gya, Titan is transiently solvent-free — a ∼100 Myr astrobiological desert when neither hydrocarbon nor water-based habitability is available.

The *Future* map (Figure 3.8) shows a near-uniform posterior mean of ≈0.44 across the entire surface — the highest global mean of any epoch. The sharp polar/equatorial contrast of the *Present* epoch is entirely replaced by the global ocean signal. Comparing with the temporal trend (Figure 3.1), this epoch represents the recovery from the solvent-free minimum at ∼+5.0 Gya; the north polar and equatorial curves converge to near-identical values, confirming the loss of spatial differentiation. The ranking is fundamentally reorganised compared with all previous epochs (Table 3.7). Cryovolcanic sites (Hotei $P(H)$=0.464; Sotra $P(H)$=0.462) lead because they combine the highest f_8 and f_5 values with maximal f_1, representing direct conduits between the ocean and accumulated organic substrate. Inundated equatorial dune fields rank 3–6 as their millennia of accumulated tholins dissolve into the global ocean. Former polar lake basins (Kraken, rank 10, $P(H)$=0.425) drop because, with f_1=1.0 everywhere, their former lake-margin advantage vanishes and they score purely on organic inventory and topography.

The score spread (0.039: from 0.425 to 0.464) is the narrowest of any epoch, reflecting the homogenising effect of the global ocean on the liquid-hydrocarbon feature — the remaining variation is driven entirely by the secondary features f_2, f_5, and f_6. The red-giant ocean window (+5.1 to +6.0 Gya, ∼0.9 Gyr) is the longest continuous period of global liquid coverage in Titan's modelled history, combining maximal organic inventory with warm water–ammonia solvent — an extraordinary future habitability scenario unmatched by any other epoch or body in the post-main-sequence solar system. Of all epochs modelled, this represents the strongest candidate for aqueous abiogenesis: warm liquid water, the full accumulated organic inventory of ∼10 Gyr of photochemistry, and a timescale (∼10^8 yr) comparable to the interval between the formation of Earth's oceans and the earliest evidence of terrestrial life. An organism that had spent billions of years perfecting cryogenic methane biochemistry would face the most dramatic environmental challenge in Titan's history — analogous to, but far more extreme than, the terrestrial Great Oxidation Event. If the transition through the solvent-free desert at ∼+5.0 Gya does not sterilise the surface entirely, the survivors would find themselves in warm water with an effectively unlimited organic buffet. With ∼0.9 Gyr of sustained molecular evolution in conditions comparable to those that produced the first eukaryotes on Earth (Knoll, 2014), the Future ocean is — at least in principle — the epoch where Titan stops being a world of patient, diffusing chemotrophs and starts being a world where biology could get ambitious.

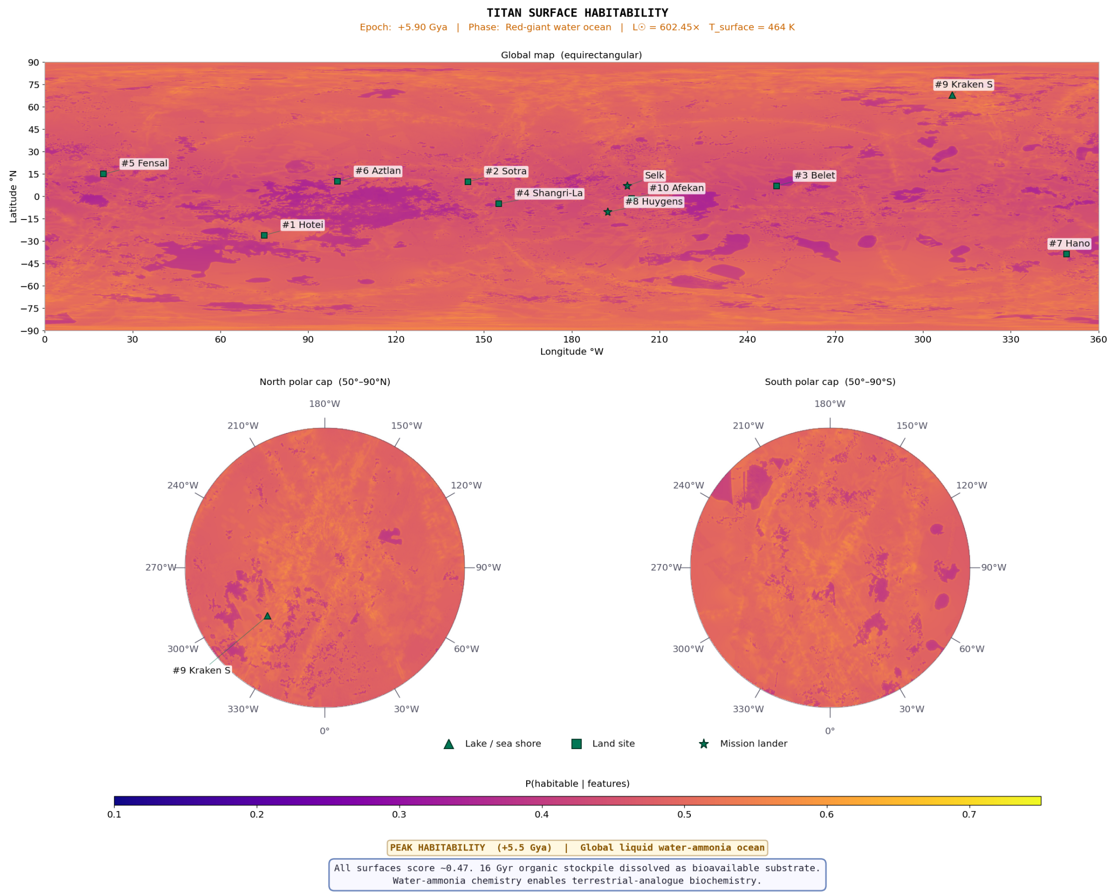

Figure 3.8: Future epoch (+5.9 Gya; red-giant ocean) posterior mean habitability. f_1 and f_8 overridden to 1.0 globally. Spatial variation reflects f_2, f_5, f_6 differences. Compare with Figure 3.4.

Table 3.7: Top-10 habitable locations at the *Future* epoch (+5.9 Gya). △ = former lake; ■ = land; ★ = lander site (Selk below line: outside top 10). $P(H)$ from Eq. 2.11.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	■ Hotei Regio	75.0	-26.0	0.464	f_1=1.0, f_8=1.0, f_2 high
2	■ Sotra Facula	144.5	+9.8	0.462	f_1=1.0, f_8=1.0, cryovol.
3	■ Belet (inundated)	250.0	+7.0	0.450	f_1=1.0, f_2 peak
4	■ Shangri-La (inund.)	155.0	-5.0	0.449	f_1=1.0, f_2 peak
5	■ Fensal (inundated)	20.0	+15.0	0.448	f_1=1.0, f_2=0.96×2.5
6	■ Aztlan (inundated)	100.0	+10.0	0.447	f_1=1.0, organics dissolved
7	■ Hano Crater	349.0	-38.6	0.441	f_1=1.0, f_8=1.0
8	★ Huygens Site	192.3	-10.6	0.436	f_1=1.0; submerged
9	■ Afekan (inundated)	200.5	-1.4	0.426	f_1=1.0
10	△ Kraken (ocean floor)	310.0	+68.0	0.425	f_1=1.0, former basin
	★ Selk Crater	199.0	+7.0	0.390	f_1=1.0; low f_2 limits rank

4 Discussion

4.1 Dragonfly Landing Site Validation

The pipeline independently validates the *Dragonfly* mission's selection of Selk crater, but through a different route than a global ranking. The maps in Chapter 3 quantify *present-day solvent* habitability; *Dragonfly* targets *past liquid-water* prebiotic chemistry. These are two scientifically distinct frameworks that correctly produce different site rankings: a model that captures present-day solvent habitability *should* rank polar lake shores above a dry equatorial crater (Appendix I). Selk's present-epoch $P(H) = 0.24$ is therefore not a contradiction of the mission rationale but a consequence of targeting different habitability.

The independent validation comes from the geomorphological diversity feature: Selk's $f_7 = 0.629$ ($7.8\times$ the global median of 0.081) is derived from Cassini SAR data alone, without reference to mission planning, and without any weight or feature specification adjusted after examining Selk. This terrain-boundary diversity — four terrain classes within 45 km — identifies the organic–water-ice contact zone that is the mission's primary scientific target.

4.2 Titan in Solar System Context

Present-epoch Titan is distinguished from all other solar system candidates by combining a confirmed cycling surface solvent with a large organic inventory. Future Titan (+5.9 Gya) uniquely pairs a water-based ocean with ∼10 Gyr of accumulated organics — the most favourable habitability combination modelled for any body in the post-main-sequence solar system.

4.3 Implications for Abiogenesis

The habitability maps produced in this thesis identify where the preconditions for life converge, but the temporal reconstruction also reveals which epochs and sites are most relevant to the *origin* of life. Two epochs stand out.

The *Past* epoch (−3.5 Gya) offers the most direct route to aqueous abiogenesis through impact-melt ponds at crater sites (Section 3.2). Liquid water, accumulated tholin substrate, atmospheric HCN, and ammonia are simultaneously present for 10^3–10^4 yr per impact — long enough for Strecker amino acid synthesis (Neish et al., 2010) but possibly too short for the transition from prebiotic chemistry to self-replicating systems. The model ranks these sites 11th and below in solvent habitability, precisely because the habitability proxy captures present-day solvent conditions rather than transient aqueous chemistry — an important caveat for interpreting the maps.

The *Future* epoch (+5.9 Gya) represents the strongest abiogenesis scenario in the entire reconstruction. The global water–ammonia ocean dissolves ∼10 Gyr of accumulated tholins, providing sustained contact between warm water solvent, complex organic substrate, and solar energy over ~0.9 Gyr — conditions broadly analogous to those invoked for early Earth but with a far richer initial organic inventory. Unlike the Archean Earth, where CO_2 and CH_4 greenhouse gases sustained liquid water under a faint Sun (Catling and Zahnle, 2020), the Future Titan ocean is maintained by direct stellar luminosity increase — a fundamentally different atmospheric context that may favour different prebiotic pathways. Unlike the transient impact-melt ponds, the timescale is comfortably long for molecular evolution: the ~0.9 Gyr ocean window is comparable to the interval between the formation of Earth's oceans and the earliest evidence of terrestrial life at ~3.5 Ga (Brasier et al., 2015). Cryovolcanic conduit sites (Hotei, Sotra) lead the ranking at this epoch because they provide direct pathways between the accumulated organic crust and the ocean interior.

The present-day polar lake shores, despite scoring highest for solvent habitability, present the most speculative abiogenesis scenario: non-aqueous abiogenesis in liquid methane at 94 K, requiring compartmentalisation via azotosome membranes (Stevenson et al., 2015) and energy from acetylenotrophy (McKay and Smith, 2005). No terrestrial analogue exists for the origin of life in a non-aqueous medium, and the low temperature severely constrains reaction kinetics. It is important to note that this framework quantifies habitability conditions — the environmental prerequisites for life — rather than biosignatures in the sense of Schwieterman and Leung (2024), which would require in-situ atmospheric or surface measurements from *Dragonfly* or a comparable mission. The habitability maps therefore highlight a fundamental tension: the sites with the highest present-day habitability scores are the least well understood in terms of abiogenesis potential, while the sites with the strongest abiogenesis credentials (impact craters, the future ocean) score modestly or are temporally inaccessible.

4.4 Limitations of the Framework

The framework relies on several simplifying assumptions, documented in full in Appendix E. The most consequential are summarised here.

Conditional feature independence. The Beta-conjugate model treats the eight features as contributing independently. In reality, f_1, f_4, and f_5 are physically coupled at lake margins. The sensitivity analysis (Section 4.5) bounds the resulting error: $|\Delta P(H)| < 0.033$, comparable to the prior-specification uncertainty.

Static geomorphological template. Feature maps are derived from 2004–2017 Cassini observations and applied uniformly to all non-present epochs via scalar scale functions. Real geological changes over billions of years — dune migration, basin infilling, tectonic modification — are unrepresented. This approximation is justified for the broad-scale habitability classification pursued here, but limits the resolution of transitional epochs.

Past-epoch Cassini contamination. The geomorphological map reflects the present surface; at the LHB, terrain distributions were different. Scale functions partially correct for this by reducing organic abundance, but the spatial pattern of f_7 is held fixed.

4.5 Sensitivity of Results to Prior Specification

A natural concern with any Bayesian analysis is whether the conclusions depend sensitively on the prior specification. Three parameters govern the prior: κ (concentration), λ (likelihood sharpness), and the feature weights w_i. Figure 4.1 shows the effect of varying each at three representative sites: Kraken (rank 1, high liquid), Belet (rank 11, high organic), and Selk (below prior mean, high geodiversity).

The key finding is robustness. Halving κ from 5 to 2.5 increases Kraken's $P(H)$ by $+0.031$ and reduces Selk's by -0.027; doubling to 10 reverses these changes. The maximum shift across all $3 \times 3 \times 3 = 27$ parameter combinations is $|\Delta P(H)| = 0.033$ — well below the 95% HDI width of ≈ 0.5 at any single site. The qualitative ranking (polar lake shores $>$ lacus $>$ equatorial organic sites) is preserved in all combinations. Crucially, reducing w_1 from 0.25 to 0.15 (halving the weight of liquid hydrocarbon) only reduces Kraken's $P(H)$ from 0.436 to 0.420 and raises Belet's from 0.295 to 0.319 — the ranking order is unchanged, confirming that the result is driven by the coincidence of multiple high-scoring features at the shoreline, not by the specific weight assigned to f_1.

Figure 4.1: Sensitivity of $P(H \mid \mathbf{f})$ to prior specification. Each row varies one parameter; each column shows one site. All deviations ≤ 0.033; qualitative rankings preserved across all 27 parameter combinations.

4.6 Future Work

The three most impactful extensions are: (1) post-*Dragonfly* recalibration of f_2 and f_8 weights using DraMS and DraGNS measurements at Selk; (2) replacing the independence assumption with a full covariance structure estimated from high-resolution SAR patches; and (3) additional temporal anchors at -2.0 to -0.5 Gya where the liquid-hydrocarbon ramp most sensitively determines whether equatorial or polar sites lead the transition.

4.7 Conclusions

This thesis presents the first quantitative, spatially-resolved, multi-epoch Bayesian habitability map of Titan's surface, derived from eight Cassini observational proxy features. Conclusions are organised by research question.

RQ1: Present-epoch spatial distribution. North polar lake margins are the most habitable present-day environments. Kraken Mare's southern shoreline achieves $P(H) \approx 0.44$ (rank 1), driven by coincident maximum f_1=1.0, f_4≈0.70, and f_7≈0.76. All ten top-10 sites are lake or lacus margins; all eight feature-weight/importance ratios lie within 10% of unity.

RQ2: LHB habitability. Cryovolcanic sites lead the *Past* ranking (Hotei $P(H)$=0.295) because their elevated f_8 and f_5 combine with the globally boosted acetylene proxy. Craters rank 11th and below in the 8-feature formula; their astrobiological significance lies in providing transient liquid water for Strecker synthesis.

RQ3: Temporal trajectory. The trajectory is non-monotonic across three regimes. (i) LHB (-3.8 to -1.5 Gya): cryovolcanic and impact-melt dominance, with tightly clustered scores reflecting the absence of a strong liquid-hydrocarbon signal. (ii) Polar-lake dominance (-1.0 to $+4.0$ Gya): lake or lacus margins occupy all top-10 positions with stable rankings across three epochs; the present-day habitability structure is a plateau extending at least 250 Myr into the future. (iii) Global ocean ($+5.1$ to $+6.0$ Gya): all sites score $P(H)>0.42$; cryovolcanic sites lead as their accumulated organic substrates dissolve into $\sim$10 Gyr of photochemical product in warm water–ammonia. A transient solvent-free minimum at $\sim$+5.0 Gya separates regimes (ii) and (iii).

RQ4: Dragonfly validation. Selk crater has the highest geodiversity of any equatorial site (f_7=0.629, 7.8× the global median), derived independently from SAR data. Its lower present-epoch score ($P(H)$=0.24) is consistent: *Dragonfly* targets past water–organic chemistry, correctly identified by the two-framework distinction (Appendix I).

Together, these results demonstrate that Titan possesses a multilayered astrobiological significance from its geological past through the present to the far future, quantifiable from the Cassini dataset with sufficient precision to guide mission planning and interpret in-situ measurements. The temporal reconstruction reveals a fundamental tension between habitability and abiogenesis (Section 4.3): the sites with the highest present-day solvent habitability scores (polar lake shores) are the least understood in terms of origin-of-life potential, while the environments with the strongest abiogenesis credentials — LHB impact-melt ponds and the red-giant ocean — score modestly or lie in Titan's temporal past and future respectively.

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A Physical Parameters for Titan

Table A.1: Physical constants and observationally-constrained parameters for Titan used throughout this thesis.

Parameter	**Value**	**Reference**
Mean radius	2575.0 km	IAU 2015
Surface gravity	$1.352\,\mathrm{m\,s^{-2}}$	Zebker et al. (2009)
Surface pressure	1.467 ± 0.001 bar	Fulchignoni et al. (2005)
Surface temperature	93.65 ± 0.25 K	Fulchignoni et al. (2005)
N_2 fraction (surface)	$\approx 94.2\%$	Niemann et al. (2005)
CH_4 fraction (surface)	$5.65 \pm 0.18\%$	Niemann et al. (2005)
Tidal Love number k_2	0.589 ± 0.150	Iess et al. (2012)
Saturn orbital period	29.46 yr	—
Axial obliquity	26.73°	—
Solar flux at Titan	$\approx 14.9\,\mathrm{W\,m^{-2}}$	at 9.54 AU
Organic deposition rate	$\sim 5 \times 10^{-14}\,\mathrm{g\,cm^{-2}\,s^{-1}}$	Lavvas et al. (2011)
North polar sea volume	$\sim 70{,}000\,\mathrm{km^3}$	Mastrogiuseppe et al. (2019)
Water–ammonia eutectic	176 K at 32.6 wt% NH_3	Grasset and Pargamin (2005)
Subsurface ocean depth	$\sim$ 55–80 km	Tobie et al. (2005)

B Bayesian Prior Specification

Table B.1: Prior means μ_i and weights w_i for all eight habitability features at the present (Cassini) epoch. Global prior mean $\mu_0 = \sum_i w_i \mu_i = 0.331$. Concentration parameter $\kappa = 5$ is applied uniformly.

Feature	**Symbol**	w_i	μ_i	**Key ref.**
Liquid hydrocarbon	f_1	0.25	0.020	Hayes et al. (2008)
Organic abundance	f_2	0.20	0.700	Malaska et al. (2025)
Chemical energy	f_3	0.20	0.350	Strobel (2010)
Methane cycle	f_4	0.15	0.400	Mitchell and Lora (2016)
Surface–atm. interaction	f_5	0.08	0.350	Miller et al. (2021)
Topographic complexity	f_6	0.06	0.250	Corlies et al. (2017)
Geomorphological diversity	f_7	0.04	0.300	Lopes et al. (2019b)
Subsurface ocean proximity	f_8	0.02	0.030	Iess et al. (2012)
Global prior mean μ_0			0.331	

C Bayesian Inference: Background and Derivation

This appendix provides the conceptual background for the Beta-conjugate Bayesian framework used in Chapter 2. Readers familiar with conjugate Bayesian updating may skip directly to the model summary in Section 2.4.

C.1 The Latent Variable

In this framework, habitability H is treated as a *latent variable* — a quantity that cannot be directly observed by any Cassini instrument but whose value is inferred from the combination of observational proxies $\mathbf{f} = (f_1, \ldots, f_8)$. A latent variable (from the Latin *latere*: to be hidden) influences observable outcomes but is not itself directly measurable. Here, H is not a binary yes-or-no property but a continuous probability in $[0, 1]$: the value $H = p$ at a given pixel means that pixel has probability p of satisfying the relevant habitability conditions given all available evidence. Each feature f_i provides partial, indirect evidence about the value of H.

C.2 Why a Beta Distribution?

The Beta distribution $\mathrm{Beta}(\alpha, \beta)$ is defined on $[0, 1]$ and characterised by two positive shape parameters α and β. Its probability density function is expressed in terms of the gamma function (Eq. 2.3 in Chapter 2):

$$p(h) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\, h^{\alpha-1}\,(1-h)^{\beta-1}, \qquad h \in [0, 1]$$

where the gamma function $\Gamma(\cdot)$ generalises the factorial to non-integer arguments (Eq. 2.4). The distribution is unimodal when both $\alpha > 1$ and $\beta > 1$, with mean $\alpha/(\alpha + \beta)$ and variance that decreases as $\alpha + \beta$ grows. Because it is bounded on $[0, 1]$ and can represent any unimodal belief over a probability, it is the natural prior for the latent habitability probability H.

C.3 Bernoulli Trials and the Binomial Likelihood

A *Bernoulli trial* is an experiment with exactly two possible outcomes — conventionally called "success" and "failure" — each occurring with fixed probabilities p and $1 - p$ respectively (DeGroot and Schervish, 2012, ch. 2). A coin flip is the prototypical example: heads (success, probability p) or tails (failure, probability $1 - p$). The *Binomial distribution* counts the total number of successes k in n independent Bernoulli trials; it is the natural likelihood function for data arising from repeated binary observations.

In the habitability model, no actual coin flips occur. Instead, the prior and data contributions are expressed as *virtual* (or *pseudo*) Bernoulli trials — a mathematical device for encoding strength of belief in the same currency as observational data. The prior contributes $\kappa = 5$ virtual trials, of which $\alpha_0 = 1.655$ are "successes" (prior evidence for habitability) and $\beta_0 = 3.345$ are "failures" (prior evidence against). Each observed feature f_i then contributes λw_i additional virtual trials, of which a fraction f_i are successes and $(1 - f_i)$ are failures. This pseudo-count interpretation is the standard Bayesian device for placing prior beliefs on the same mathematical footing as observed data (Gelman et al., 2013, ch. 2), and is the mechanism by which conjugacy yields a closed-form posterior (Section C.4).

C.4 Conjugacy

A prior distribution is *conjugate* to a likelihood function if the posterior (prior updated by data) has the same mathematical form as the prior, differing only in its parameter values (Gelman et al., 2013, ch. 2). The Beta distribution is conjugate to the Binomial likelihood: if the prior on a success probability p is $\text{Beta}(\alpha_0, \beta_0)$ and we observe k successes in n Bernoulli trials, the posterior is $\text{Beta}(\alpha_0 + k,\ \beta_0 + (n - k))$. In the habitability model, this update takes the specific form given in Eqs. 2.7–2.8, where the pseudo-counts $k = \lambda \sum_i w_i f_i$ and $n - k = \lambda \sum_i w_i (1 - f_i)$ are derived from the weighted feature values. The resulting posterior PDF (Eq. 2.9) has the same gamma-function structure as the prior PDF (Eq. 2.3), with only the shape parameters changed — no new functional form is introduced. This analytical tractability makes the update feasible at 6.49 million pixels simultaneously with no numerical integration required.

C.5 The Beta-Binomial Model Applied to Habitability

Each feature $f_i \in [0, 1]$ is treated as the fractional outcome of λw_i virtual Bernoulli trials: fraction f_i of those trials are "successes" (evidence for habitability) and $(1 - f_i)$ are "failures" (evidence against). High feature values contribute more positive pseudo-observations and shift the posterior towards higher habitability; low values do the opposite. The prior contributes κ virtual observations and the data from all eight features contribute a further $\lambda \sum_i w_i = \lambda$ observations.

Figure 2.3 visualises the updating process for three representative sites (Kraken southern shore, Belet dune field, Selk crater), showing how each successive feature narrows and displaces the Beta distribution from the prior to the final posterior. The figure also shows the 95% HDI for eight key present-epoch sites.

Figure 2.2 illustrates this update graphically for four representative sites. The left panel shows the prior $\text{Beta}(\alpha_0, \beta_0)$, which is weakly informative ($\sigma \approx 0.19$) and centred at $\mu_0 = 0.331$. The right panel shows the posteriors after incorporating the respective Cassini feature values: the Kraken Mare shoreline and Ligeia shoreline shift rightward (high liquid, methane-cycle, and geodiversity scores), while the Belet dune sea shifts only modestly (high organic but no liquid) and Selk crater shifts leftward (below-average methane cycle and surface–atmosphere scores despite exceptional geodiversity).

D Geomorphological Organic Abundance Scores

Table D.1: Organic abundance proxy scores used for the geomorphological seamless mode of Feature 2. Grounded in published VIMS spectral characterisation of each terrain class.

Class	Name	Physical basis	Score
Dunes	Equatorial sand seas	Tholin-dominated; highest VIMS ratio (Lorenz et al., 2006)	0.82
Plains	Smooth lowlands	Organic sediment blanket (Lopes et al., 2019a)	0.68
Basins	Topographic lows	Organic accumulation (Lopes et al., 2020)	0.58
Labyrinth	Dissected terrain	Spectral affinity with plains (Malaska et al., 2020)	0.55
Craters	Impact structures	Mixed water-ice and organic ejecta (Neish et al., 2018)	0.35
Mountains	Ridge terrain	Water-ice-rich crustal material (Solomonidou et al., 2018)	0.25
Lakes	Hydrocarbon seas	Liquid hydrocarbons; low solid organic content (Brown et al., 2008)	0.05

E Declared Assumptions and Known Limitations

A1. Conditional feature independence. The eight features are treated as conditionally independent given habitability H. In practice, organic abundance and chemical energy are positively correlated, as are liquid hydrocarbon and surface–atmosphere interaction. The independence assumption is conservative: correlated positive features slightly overstate the posterior, but the framework remains physically motivated and the data-derived importances are consistent with the prior weights (Section 3.4.3).

A2. Static geomorphological template. The Lopes et al. (2019) map is used for all five temporal configurations. The Late Heavy Bombardment surface had different dune distributions and crater density, and the future surface will be submerged. The present template is used as the best available proxy, with amplitude modifications applied through epoch-specific prior means and scale functions.

A3. Seamless organic abundance mode. The organic abundance feature uses terrain-class scores globally rather than VIMS spectral band ratios, because the VIMS mosaic covers only $\sim$50% of the globe and direct blending produced a visible seam at the coverage boundary at 180°W.

A4. Past-epoch Cassini feature contamination. The past-epoch posterior was computed using present-era Cassini SAR maps, which encode the present polar lake morphology even at $t = -3.5$ Gya. The north polar Bayesian score in the past anchor is inflated above the physically expected prior value. This is mitigated in the multi-epoch reconstruction by using the analytical Bayesian formula directly for pre-lake-formation frames (Section 2.6) and is documented in Section 4.4.

A5. Multi-epoch temporal scaling accuracy. The multi-epoch reconstruction applies scalar scale functions $s_i(t)$ to the present-epoch feature maps and re-evaluates the Bayesian posterior formula at each epoch (Section 2.6.1). This is physically motivated but is not equivalent to independent epoch-specific feature extraction. The scale functions are calibrated to reproduce known physical transitions (lake formation, solar warming, eutectic crossing) but have no observational constraint at intermediate epochs. Estimated uncertainty: $\Delta P(H) \lesssim \pm 0.05$ at non-anchor epochs.

A6. SAR coverage gap. Approximately 33% of Titan's surface lacks SAR coverage. Features 1, 3, and 8 are degraded in these regions, where geomorphological and topographic proxies substitute.

F Acronyms, Notation, and Data Source Details

F.1 Acronyms and Nomenclature

The following acronyms are used throughout this chapter and the remainder of the thesis. They are defined here for convenience and also at the point of first use.

SAR Synthetic Aperture Radar. An active microwave imaging technique in which the motion of the spacecraft is used to synthesise a very long antenna aperture, yielding fine-resolution imagery of surface texture and composition.

VIMS Visible and Infrared Mapping Spectrometer. The *Cassini* instrument that imaged Titan's surface through six near-infrared atmospheric transparency windows, providing surface composition data.

CIRS Composite Infrared Spectrometer. The *Cassini* thermal emission instrument used here to constrain the surface temperature field.

GTDR Global Titan Digital elevation model, Radar — the raw Cassini RADAR altimetry dataset archived in the Planetary Data System.

GTDE Global Titan Digital Elevation model — the globally interpolated topographic product of Corlies et al. (2017) derived from the GTDR and supplementary data sources.

DEM Digital Elevation Model. A gridded representation of surface topography.

PDS Planetary Data System. The NASA archive for planetary mission data.

HDI Highest Density Interval. A Bayesian credible interval containing the specified probability mass within the narrowest possible range.

F.2 Notation

Two quantities appear throughout this chapter and are defined here to avoid repetition. Each habitability-proxy feature f_i ($i = 1, \dots, 8$) is assigned:

- A **weight** $w_i \in [0, 1]$, where $\sum_i w_i = 1$. The weight encodes the relative astrobiological importance of feature i as a habitability indicator. A feature with a higher weight exerts a larger influence on the posterior probability of habitability, all else being equal. Weights are assigned from physical reasoning and validated against data-derived importances (Section 3.4.3).

- A **prior mean** $\mu_i \in [0, 1]$. This is the expected spatially-averaged value of feature i before any observational data are considered — a statement of our prior knowledge about how common or widespread the habitability-relevant condition is on Titan. For example, liquid hydrocarbon covers ~2% of Titan's surface (Hayes et al., 2008), so its prior mean is $\mu_1 = 0.02$. The weighted sum of all prior means defines the global prior mean habitability: $\mu_0 = \sum_i w_i \mu_i = 0.331$. Note that the value $\mu_0 = 0.331$ does not appear in the prior literature as a global habitability estimate for Titan; it emerges from the component prior means in Table B.1 and should be interpreted as encoding the observation that Titan's surface is globally organic-rich whilst simultaneously recognising that liquid solvent is present at only ~2% of the surface. The resulting moderate prior is intentionally revisionable: the weak concentration parameter $\kappa = 5$ ensures that the data, not the prior, determine the posterior at any pixel where observational coverage is good. Full values are tabulated in Appendix B.

This notation first appears in a practical context in Section 2.3, where the eight features are described individually. Their role in the Bayesian update is explained in Section 2.4.

F.3 Observational Data Source Details

F.3.1 Cassini RADAR Synthetic Aperture Radar Mosaic

The Cassini RADAR instrument operated in SAR mode during Titan flybys, imaging the surface at ~350 m per pixel over the 127 targeted flybys from 2004 to 2017 (Janssen et al., 2016). SAR measures the normalised radar cross-section σ^0 (typically expressed in dB), which is sensitive to surface roughness, dielectric properties, and the presence of smooth or conducting materials. Low backscatter ($\sigma^0 < -20$ dB) is diagnostic of smooth surfaces (consistent with liquid hydrocarbon lakes) or dielectrically absorbing material (consistent with tholin-rich dune sand). High backscatter correlates with rough, water-ice-rich material such as mountain terrain, crater ejecta, and some cryovolcanic surfaces. The SAR mosaic covers ~67% of Titan's surface and constitutes the primary data source for the liquid hydrocarbon, chemical energy, and subsurface ocean features.

F.3.2 VIMS Near-Infrared Spectral Mosaic

The Cassini Visible and Infrared Mapping Spectrometer (VIMS) observed Titan's surface through atmospheric transparency windows at 0.94, 1.08, 1.27, 1.59, 2.03, and 2.79 µm. The band ratio 1.59/1.27 µm is an established proxy for tholin (complex organic) abundance at the surface: the 1.59 µm window probes deeper into the tholin-bearing surface whilst the 1.27 µm window provides atmospheric normalisation (Barnes et al., 2007). The Seignovert et al. (2019) combined VIMS+ISS mosaic (Seignovert et al., 2019) provides this band ratio map at 2–4 km per pixel coverage, but only for the 0–180°W hemisphere (~50% of the globe). The significance of this coverage gap for the organic abundance feature is discussed in Section G.2.

F.3.3 CIRS Thermal Emission Observations

The Composite Infrared Spectrometer (CIRS) measured Titan's surface brightness temperature at far-infrared wavelengths throughout the Cassini mission. Jennings et al. (2019) derived an analytical model of the latitude-dependent surface temperature from 13 years of CIRS observations, producing the formula:

$$T(\phi, Y) = (93.53 - 0.095\,Y)\cos\Big[(\phi + 0.85 - 3.2\,Y)(0.0029 - 0.00006\,Y)\Big] \tag{F.1}$$

where ϕ is latitude in degrees and $Y = t_{\mathrm{CE}} - 2009.61$ is years from the northern spring equinox (22 May 2009). This formula reproduces the observed zonal surface temperatures to within ± 0.4 K (rms) across the full Cassini epoch and provides 100% global coverage, grounded in CIRS focal-plane-1 limb and nadir observations across the full 2004–2017 mission. The model captures only latitudinal and seasonal variation; no longitudinal inhomogeneity is represented (consistent with the absence of significant longitudinal temperature gradients in the Cassini dataset). For temporal epochs other than the present, the mean annual temperature field is held stationary: this approximation is justified because the dominant habitability driver at non-present epochs is the liquid-hydrocarbon scale function $s_1(t)$ rather than the temperature gradient, and the methane-cycle feature contributes only $w_4 = 0.15$ to the posterior. It is used to synthesise the global temperature field and its meridional gradient, which enters the methane cycle feature (Section G.4).

F.3.4 GTDE/GTDR Topographic Models

The Global Titan Digital Elevation model (GTDE), produced by Corlies et al. (2017), integrates Cassini RADAR altimetry, SARtopo stereophotogrammetry, and radial basis function interpolation to produce a globally-interpolated topographic model at ~22 km per pixel (2 pixels per degree). Coverage is approximately 90% of the globe, with the south polar region and some high-latitude northern areas relying most heavily on interpolation. The topographic model contributes to the chemical energy feature (identifying topographic lows as preferential organic accumulation sites), the topographic complexity feature (through local roughness estimation), and the surface–atmosphere interaction feature (through slope calculation).

F.3.5 Lopes (2019) Global Geomorphological Map

The global geomorphological map of Lopes et al. (2019b) classifies the Cassini SAR-imaged surface into seven terrain units (plains, dunes, mountains, basins, labyrinth terrain, craters, and lakes) using SAR texture, backscatter, morphology, and VIMS spectral correlation. This map provides the terrain classification that enters the organic abundance feature (as a globally seamless gap-fill) and the geomorphological diversity feature. The shapefiles are provided in east-positive geographic coordinates and require conversion to the west-positive convention of the analysis grid.

F.3.6 Miller (2021) Global Fluvial Channel Map

Miller et al. (2021) identified and mapped Titan's global fluvial channel network from Cassini SAR data, producing a vector dataset of channel locations and widths. This network constitutes the primary mechanism for organic material transport from equatorial source regions to the polar basins, and channel density is used as one component of the surface–atmosphere interaction feature.

F.3.7 Tidal Love Number and Gravity Data

The degree-2 tidal Love number $k_2 = 0.589 \pm 0.150$ from Iess et al. (2012) provides a global constraint on interior structure, confirming the presence of a subsurface liquid layer. This scalar quantity enters as a global floor value in the subsurface ocean proximity feature.

F.3.8 Hedgepeth (2020) Global Crater Catalogue

The global catalogue of confirmed and probable impact structures on Titan by Hedgepeth et al. (2020) records crater diameters and positions for all resolved impact features in the Cassini SAR dataset. This catalogue is used exclusively in the *Past* epoch to generate the impact-melt proximity feature.

F.4 Projection and Reprojection

Three coordinate reference systems are used throughout the pipeline; this section documents each and the transformations between them.

F.4.1 Canonical Equirectangular Grid

All analysis is performed on a simple cylindrical (equirectangular) grid with west-positive longitude (0–360°W) and latitude ±90°. The grid dimensions are 1802×3603 pixels at 4490 m/px. This projection preserves longitude–latitude coordinates at the cost of area distortion at the poles; it is the native CRS of the Cassini RADAR and GTDR raster products distributed by USGS Astropedia.

F.4.2 GeoTIFF Reprojection via Rasterio

Raw Cassini data products are delivered as GeoTIFFs in a projected coordinate system (coordinates in metres, SimpleCylindrical CRS). These are reprojected onto the canonical grid using `rasterio.warp.reproject` with bilinear resampling. The reprojection handles the coordinate transformation from the source CRS embedded in each GeoTIFF to the canonical grid's CRS, nodata masking (source nodata values, typically 0.0, are replaced with NaN before reprojection), and resampling from the source resolution to the canonical 4490 m/px grid. This is the only stage at which a GIS reprojection library is used; all subsequent analysis operates in the canonical equirectangular coordinate system.

F.4.3 Polar Stereographic Insets

The polar inset maps in the animation frames and temporal reconstruction figures use an oblique polar stereographic projection following Snyder (1987). The projection maps disc coordinates (x, y) with $r = \sqrt{x^2 + y^2} \in [0, 1]$ to geographic coordinates via:

$$\phi = 90 - 2 \arctan\bigl(r \cdot k_{\mathrm{PS}}\bigr), \qquad \lambda = \arctan 2(x,\, y) \tag{F.2}$$

where ϕ is latitude, λ is west longitude, and $k_{\mathrm{PS}} = \tan\bigl((90 - \phi_{\mathrm{edge}})/2\bigr)$ is a pre-computed scale factor that maps the disc edge ($r = 1$) to the desired polar cap boundary ϕ_{edge} (55° by default). For the south polar disc, ϕ is negated. Pixel values are obtained by bilinear interpolation from the equirectangular grid using the inverse-projected latitude and longitude at each disc pixel. This hand-coded implementation avoids introducing a dependency on cartographic projection libraries for what is purely a visualisation transformation.

G Feature Engineering: Scientific Basis and Implementation

The eight habitability-proxy features are extracted from the Cassini data sources described in Appendix F. Each subsection gives the scientific basis, prior mean justification, implementation procedure, and missing-data backfill strategy.

Eight habitability-proxy features are extracted from the above data sources. Figure G.1 shows all eight feature maps at the present epoch, illustrating their distinct spatial patterns. Figure G.2 shows how the relative activity of each feature is modulated across geological time.

All features are normalised to the range $[0, 1]$ using a robust percentile-clipping approach:

$$\hat{f}_i = \text{clip}\left(\frac{f_i - f_{i,\text{p2}}}{f_{i,\text{p98}} - f_{i,\text{p2}}},\ 0,\ 1\right) \tag{G.1}$$

where $f_{i,\text{p2}}$ and $f_{i,\text{p98}}$ are the 2nd and 98th percentiles of finite values in the global layer, mapping the central 96% of the observed dynamic range onto $[0, 1]$ whilst preventing extreme outliers from compressing the scale. Where a feature is unavailable at a particular pixel (due to data gaps), the value is set to NaN and handled through the Bayesian missing-data imputation procedure described in Section 2.4.3.

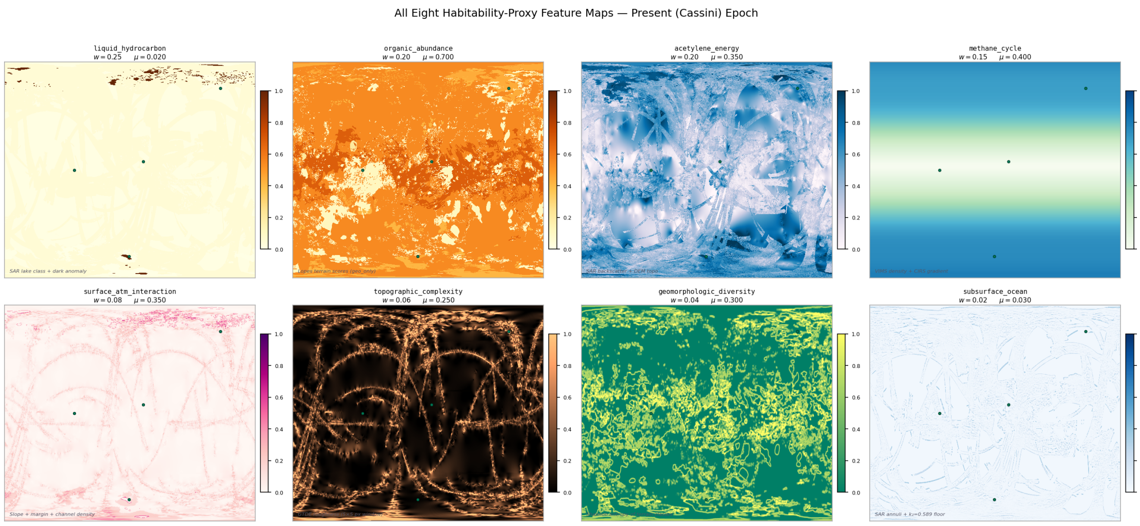

Figure G.1: All eight habitability-proxy feature maps at the present (Cassini) epoch, each normalised to $[0, 1]$. The panels illustrate the distinct spatial patterns contributed by each feature: the polar concentration of liquid hydrocarbon (f_1), the near-global organic abundance (f_2), the topographically-modulated chemical energy proxy (f_3), and the latitude-dependent methane cycle signal (f_4), amongst others. The combined posterior is derived from all eight simultaneously via the Bayesian update described in Section 2.4. Prior weight w_i and prior mean μ_i are shown for each panel.

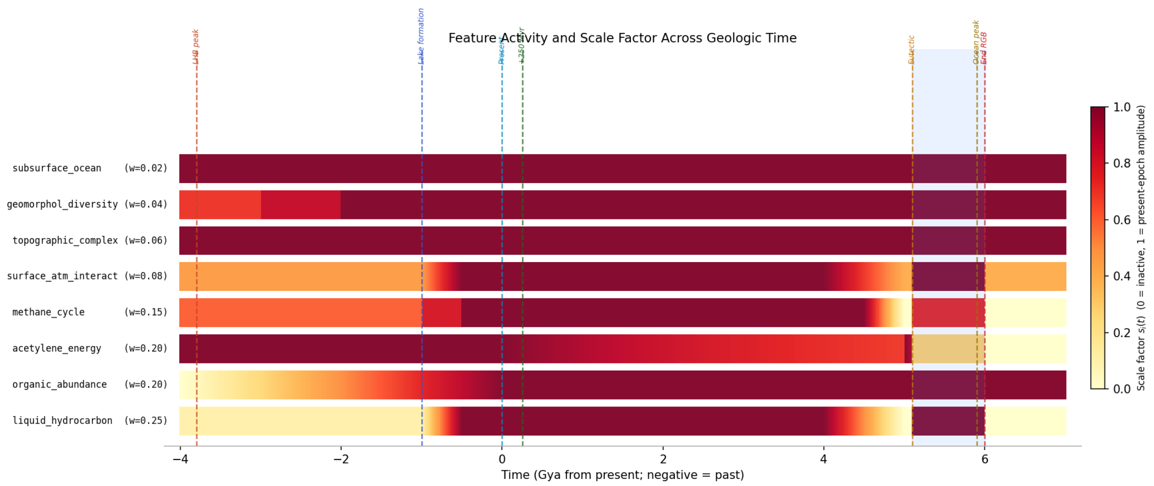

Figure G.2: Scale factor (relative activity) for each feature as a function of geological time from −4.0 Gya to +7.0 Gya. Colour intensity encodes the scale applied to each feature's prior mean at that epoch: bright yellow indicates full present-epoch amplitude; dark blue indicates near-zero activity. Key events (LHB peak, lake-formation onset, present epoch, solar warming ramp, eutectic crossing, ocean peak) are marked with vertical dashed lines. The impact-melt proxy and cryovolcanic flux features are active only in pre-lake epochs and are shown in the lower two rows.

G.1 Feature 1: Liquid Hydrocarbon Availability ($w_1 = 0.25, \mu_1 = 0.02$)

Scientific basis. Liquid methane and ethane are the proposed non-aqueous biochemical solvents for any life operating on Titan's present-day surface (McKay, 2016; McKay and Smith, 2005). At the ambient surface temperature of ∼94 K, water is structurally as rigid as rock and cannot support molecular diffusion processes fundamental to metabolism. Liquid methane and ethane are, however, stable at 94 K and 1.47 bar, as confirmed by the permanent north polar seas. Benner et al. (2004) proposed that non-polar solvents can in principle support biochemistry through different but self-consistent molecular architectures, and Stevenson et al. (2015) demonstrated through molecular dynamics simulations that acrylonitrile molecules present in Titan's atmosphere can form stable vesicle membranes in liquid methane. The presence of persistent liquid hydrocarbon is therefore treated as the single most important habitability prerequisite for surface life at the present epoch, and accordingly receives the highest prior weight ($w_1 = 0.25$).

Prior mean. The north polar seas cover approximately 1.5–2% of Titan's surface (Hayes et al., 2008), giving a prior mean of $\mu_1 = 0.02$. Despite the high weight, the low prior mean means that most pixels contribute negligibly through this feature; its habitability impact is strongly localised to the polar regions.

Implementation. The primary source is the Lopes et al. (2019) lake polygon class (terrain class 7), rasterised to the canonical grid as a binary lake mask. A secondary contribution comes from SAR backscatter dark anomalies not otherwise attributed to organic material. Confirmed empty lake basins from the Birch et al. (2017) analysis are explicitly assigned

$f_1 = 0$ to reflect the absence of present-day liquid even within geomorphologically defined basins. In the absence of SAR coverage, ISS 938 nm albedo imagery provides a proxy, with smooth, low-albedo regions consistent with liquid surfaces elevated towards higher feature values.

Backfill. In regions lacking both SAR and ISS coverage, the feature is assigned NaN and imputed at the prior mean during inference.

G.2 Feature 2: Organic Abundance ($w_2 = 0.20$, $\mu_2 = 0.70$)

Scientific basis. Tholins — the complex refractory macromolecular organic compounds produced continuously by ultraviolet photolysis of the N_2/CH_4 mixture in the upper atmosphere — are the dominant surface material across most of Titan. Their astrobiological significance is multifaceted. Laboratory hydrolysis of tholins in liquid water at biologically relevant temperatures (273–373 K) produces amino acids, nucleobase analogues, and sugar-like polyols at yields of ∼0.1% by mass (Neish et al., 2018), directly confirming that tholins can serve as a chemical feedstock for prebiotic synthesis when liquid water is available — either from impact melts at past epochs or from the red-giant ocean in the far future. Meyer-Dombard et al. (2025) characterise the tholin layer as the primary chemical substrate for any surface habitability scenario on Titan. The feature weight ($w_2 = 0.20$) is second only to liquid hydrocarbon, reflecting this central role.

Prior mean. The revised prior mean of $\mu_2 = 0.70$ (updated from 0.60 in earlier pipeline versions) reflects updated spectral characterisation from Malaska et al. (2025) and Cable et al. (2012), indicating that approximately 82% of the SAR-classified surface by area falls into organic-rich terrain classes (dune: 17%, plains: 65%). The organic abundance feature consequently makes the largest single contribution to the global prior mean ($w_2\mu_2 = 0.140$).

Implementation and the seamless organic mode. The primary spectral data source is the VIMS+ISS 1.59/1.27 µm band-ratio mosaic from Seignovert et al. (2019), but this covers only the 0–180°W hemisphere (∼50% of the globe). Extensive testing of direct blending between the VIMS band ratios and geomorphological class scores in the 180–360°W hemisphere produced a visible seam artefact at the 180°W boundary: the absolute calibration of the VIMS spectral ratio differs systematically from the terrain-class score at the coverage boundary, producing a step change of ∼0.13 in median organic abundance clearly visible in rendered maps. This artefact cannot be eliminated by a single global offset because the boundary region terrain is not representative of all terrain classes. The analysis therefore uses a seamless geomorphological mode, applying the terrain-class scores from Table D.1 (Appendix D) globally. Whilst this approach sacrifices per-pixel spectral variation within the VIMS-covered hemisphere (reducing the standard deviation from 0.27 for VIMS data to 0.16 for terrain-class scores), it ensures that no spatial artefact from the uneven VIMS coverage contaminates the habitability maps.

Backfill. The Lopes (2019) terrain classification covers ~90% of the SAR-imaged surface (~67% of the globe). In regions outside SAR coverage, terrain class is estimated from ISS albedo units with reduced specificity.

G.3 Feature 3: Chemical Energy Availability ($w_3 = 0.20$, $\mu_3 = 0.35$)

Scientific basis. Life requires a sustained source of chemical free energy to drive metabolic reactions against thermodynamic equilibrium. On Titan's surface, the primary candidate is acetylenotrophy ($C_2H_2 + 3\,H_2 \longrightarrow 2\,CH_4$, $\Delta G = -334\,\mathrm{kJ\,mol^{-1}}$; where ΔG is the Gibbs Free Energy change of the reaction, a negative value indicating spontaneous energy release) available from acetylene–hydrogen combination (McKay and Smith, 2005). Yanez et al. (2024) recalculated the available energy under updated thermochemical conditions at Titan's surface as 69–78 $\mathrm{kJ\,mol^{-1}}$ C, well above the minimum for terrestrial methanogens. Acetylene is produced continuously by UV photolysis of methane and sediments to the surface where it has been directly detected by GCMS (Niemann et al., 2005). The chemical energy feature is therefore a proxy for the co-availability of acetylene and hydrogen at the surface.

Implementation. Since the acetylene surface abundance cannot be directly mapped by any Cassini instrument, it is approximated through two components. SAR backscatter (weight 70% where coverage exists): SAR-dark surfaces characteristic of organic-rich terrain represent regions where acetylene is most likely to accumulate without competing abiotic reactions consuming it rapidly. Topographic inversion (weight 30%): acetylene and other atmospheric condensates preferentially accumulate in topographic lows through both gravitational settling and fluvial transport; the normalised negative of the GTDE elevation ($-z$) assigns high values to topographic lows. In SAR-absent regions, the feature reverts to the pure topographic inversion component.

G.4 Feature 4: Methane Cycle Intensity ($w_4 = 0.15$, $\mu_4 = 0.40$)

Scientific basis. The methane cycle is the dominant agent of chemical transport on Titan. It delivers organic material from equatorial photochemical production regions to the polar lake margins, maintains chemical gradients at shorelines, and provides seasonal wet–dry cycling at lake edges that some prebiotic chemistry scenarios identify as important for concentrating organic reactants and driving condensation reactions (Benner et al., 2004; Mitchell and Lora, 2016). Regions experiencing the most active methane cycling receive fresh organic inputs most regularly and maintain the most dynamic chemical environments.

Implementation. The methane cycle intensity is a weighted composite of three independently computed components:

1. **VIMS observation density** (50%): the number of VIMS spectral footprints per grid cell, smoothed with a 5-pixel Gaussian kernel. Cassini's VIMS observation tar-

geting was scientifically motivated, prioritising regions of active methane-cycle interest; higher coverage density therefore correlates with active methane cycling zones (Le Mouélic et al., 2019).

2. **Meridional temperature gradient** $|\partial T/\partial \phi|$ (25%): derived from the Jennings (2019) model (Eq. F.1), evaluated at $Y = +1.39$ (year 2011.0, the approximate mission midpoint). Regions where the surface temperature gradient is steepest experience the strongest differential evaporation and precipitation, driving the most vigorous methane transport.

3. **Latitude prior** (25%): the function $\exp[-(\phi \pm 45)^2/(25)^2]$, following Mitchell and Lora (2016)'s general circulation modelling result that the mid-latitude regions host the primary methane cycle activity during the Cassini epoch.

This feature has 100% global coverage by construction and requires no backfill.

G.5 Feature 5: Surface–Atmosphere Interaction Intensity ($w_5 = 0.08$, $\mu_5 = 0.35$)

Scientific basis. The physical interface between surface and atmosphere is where gas-phase photochemical products encounter potential liquid solvents and organic substrates. Lake margins and fluvial channel termini are the primary loci of this exchange. Hayes (2016) describes the complex sedimentological and chemical processes at hydrocarbon lake shorelines, including the evaporation-driven concentration of dissolved organics and the seasonal flooding and drying cycles that expose organic sediments to liquid contact. Miller et al. (2021) documents that the fluvial channel network terminates preferentially at polar lake margins, establishing these as zones of maximum organic delivery and chemical gradient activity.

Implementation. A weighted sum of three spatial components:

$$f_5 = 0.30\,\hat{s} + 0.40\,\hat{m} + 0.30\,\hat{c} \tag{G.2}$$

where $\hat{s}$ is the normalised topographic slope $|\nabla z|$ from the GTDE DEM (high slopes drive physical transport and mixing); $\hat{m}$ is a binary lake-margin mask extended to $\sim$45 km from the nearest confirmed lake shore; and $\hat{c}$ is the Gaussian-smoothed ($\sigma = 3$ pixels) channel density from the Miller et al. (2021) dataset, normalised at the 2nd–98th percentile range. The lake-margin component receives the highest within-feature weight (40%) because shorelines concentrate all three relevant processes — liquid contact, organic delivery, and thermal/chemical cycling.

G.6 Feature 6: Topographic Complexity ($w_6 = 0.06$, $\mu_6 = 0.25$)

Scientific basis. Local topographic roughness at $\sim$20 km scale correlates with the dissected terrain types that most frequently expose water-ice-bearing crustal material (Lopes et al., 2019b). On Earth, topographic heterogeneity at ecologically relevant scales creates micro-environmental diversity — variations in drainage, aspect, moisture, and chemical exposure — that underpins elevated species richness at landscape boundaries. On Titan, the analogous principle is that varied topography produces diverse chemical micro-environments with potentially richer reactive diversity.

Implementation. Local standard deviation of GTDE elevation computed in a 5×5 pixel ($\sim$22 km) sliding window, normalised to $[0, 1]$ at the 2nd–98th percentile range. South polar gaps in the primary GTDE are filled with the Corlies et al. (2017) globally interpolated DEM prior to roughness computation.

G.7 Feature 7: Geomorphological Diversity ($w_7 = 0.04$, $\mu_7 = 0.30$)

Scientific basis. Astrobiologically relevant chemistry is disproportionately likely to occur at interfaces between geochemically distinct terrain types. The reasoning is thermodynamic: chemical gradients — differences in redox potential, organic content, water-ice availability, or pH — drive reactions that would not proceed in chemically uniform settings (Shock and Holland, 2007). On Titan, the most important geochemical interfaces identified from Cassini data are:

1. The **dune–plains boundary**, where organic-rich tholin-saturated dune sand is in direct lateral contact with the moderate-organic plains through which fluvial channels transport material towards the polar lakes. This interface concentrates the exchange between the primary photochemical organic source (dunes) and the active transport system (plains channels).

2. The **crater ejecta interface**, where water-ice-rich crustal material excavated by impact is juxtaposed with adjacent organic-bearing terrain. Neish et al. (2018) demonstrated experimentally that mixing water ice and tholins under liquid water conditions produces amino acids; the spatial co-occurrence of these two materials at crater ejecta boundaries is therefore a direct proxy for the relevant chemical reactivity.

3. The **lake–plains/dune boundary**, where liquid hydrocarbon meets the organic substrate, enabling dissolution and transport of surface organics into the lake system.

The Shannon entropy of terrain class distribution within a local window quantifies the degree of terrain-type mixing at each pixel. A pixel surrounded by a single terrain class (entropy $= 0$) sits in a geochemically homogeneous environment. A pixel at the intersection of four terrain classes (entropy approaching $H_{\max}$) sits at a multi-way geochemical interface. This feature receives the lowest weight of the present-epoch set ($w_7 = 0.04$) because it is an indirect and geometrical proxy for chemical gradients rather than a direct measurement.

Implementation. The Shannon entropy of the terrain class distribution in a sliding window of radius 10 pixels (∼45 km), normalised by the theoretical maximum entropy for the number of classes present:

$$f_7(r,c) = \frac{H(r,c)}{H_{\max}}, \qquad H(r,c) = -\sum_{k=1}^{N_c} p_k(r,c) \ln p_k(r,c), \qquad H_{\max} = \ln N_c \tag{G.3}$$

where $p_k(r,c)$ is the fraction of window pixels classified as terrain class k, and N_c is the number of distinct classes present in the window. Shannon entropy is the standard information-theoretic measure of diversity (Shannon, 1948) and has been applied to quantify geomorphological and ecological diversity in planetary and terrestrial settings (Lorenz, 2014). The normalisation by $H_{\max} = \ln N_c$ maps the maximum possible entropy (uniform distribution across all present classes) to 1.0, ensuring the feature is comparable across pixels with different numbers of terrain classes.

G.8 Feature 8: Subsurface Ocean Proximity ($w_8 = 0.02$, $\mu_8 = 0.03$)

Scientific basis. Titan's confirmed subsurface water–ammonia ocean (Iess et al., 2012) is potentially habitable by conventional criteria (Affholder et al., 2025), and there is indirect evidence that material from this ocean has reached the surface at specific locations. SAR-bright annular features surrounding several impact craters are most consistently interpreted as ancient impact-melt or cryolava fronts where liquid spread outward from basin centres, potentially establishing chemical contact between the subsurface ocean and the organic surface inventory (Wood et al., 2010). The prior mean is set at $\mu_8 = 0.03$ (revised downward from 0.10 following Neish et al. 2024), reflecting the genuine uncertainty in the surface expression of the subsurface ocean at the present epoch.

Implementation. The feature is constructed in two parts. First, a global floor value of $p_{\text{floor}} = 0.03$ is applied uniformly to all pixels, reflecting the global presence of the subsurface ocean as confirmed by the tidal Love number k_2 (Iess et al., 2012). This floor captures the fact that the ocean is a global structure and that the possibility of ocean–surface exchange cannot be excluded at any location, even if no surface expression is visible.

Second, localised enhancements above the floor are applied at SAR-bright annular structures identified in the Cassini RADAR mosaic. These annuli are selected by their backscatter ($\sigma^0 > 0$ dB, indicating rough or water-ice-rich material), their plan-view morphology (roughly circular or lobate, distinct from the linear patterns of mountain ridges or the radar-dark texture of dunes), and their association with confirmed or probable impact structures in the Hedgepeth et al. (2020) catalogue. At each such structure, a two-dimensional Gaussian kernel of half-width $\sigma \approx 200$ km centred on the crater creates a smooth spatial enhancement. The 200 km scale is set to approximate the spatial extent of the ancient melt-water or cryolava deposit that the annulus represents: for a 90 km diameter crater (Selk), melt-front runout distances of ∼200–500 km are consistent with the low-viscosity water–ammonia melt and the low-gravity environment (O'Brien et al., 2005).

The feature is the normalised sum of the global floor and all Gaussian kernels, clipped to $[0, 1]$. Because the global floor is already low ($\mu_8 = 0.03$) and only a small number of confirmed annuli exist in the SAR mosaic, the spatial variance of f_8 is very low: it is 0.03 everywhere except within $\sim$500 km of a confirmed annular structure, where it reaches values of 0.15–0.25.

H The Cassini–Huygens Mission and Dragonfly

The *Cassini–Huygens* mission (2004–2017) provided the observational foundation of this thesis through 127 targeted Titan flybys, the *Huygens* atmospheric probe descent and landing, and a systematic multi-instrument survey of Titan's surface, atmosphere, and interior. SAR mapping covered $\sim$67% of the surface; VIMS near-infrared spectroscopy $\sim$50%; CIRS thermal mapping the full globe at $\sim$10° resolution. The mission's key unresolved gap is the absence of a quantitative, spatially-resolved, multi-epoch habitability analysis integrating all data modalities — which this thesis provides.

NASA's *Dragonfly* rotorcraft (Lorenz et al., 2018), due to arrive in 2034, will traverse $\sim$175 km from the Belet dune field to Selk crater over a 2.7-year mission. Chapter 3 provides a quantitative evaluation of that site selection relative to globally optimal alternatives.

I Dragonfly Mission: Site Validation and Instrument Mapping

I.1 Mission Science Objectives

NASA's *Dragonfly* rotorcraft (Lorenz et al., 2018), due at Titan in 2034, traverses ∼175 km from Belet dunes to Selk crater. Its instruments — DraMS (mass spectrometry), DraGNS (gamma-ray and neutron spectrometry), DragonCam (imaging), and DraGMet (meteorology) — directly measure the features underpinning the habitability scores at its landing sites.

I.2 Two Distinct Habitability Frameworks

Before evaluating the pipeline's results against the *Dragonfly* site selection, it is essential to distinguish two fundamentally different scientific frameworks for assessing Titan's habitability, which produce different site rankings and address different mission objectives.

Present-day solvent habitability. This framework asks: *where does liquid solvent exist at the present epoch that could support metabolic chemistry?* For water-based life, the answer is the inaccessible subsurface ocean. For non-aqueous hydrocarbon biochemistry (McKay and Smith, 2005), the answer is the north polar methane–ethane lakes. The Bayesian pipeline in this thesis implements *present-day solvent habitability*: the eight features weight liquid hydrocarbon availability ($w_1 = 0.25$), methane cycle activity ($w_4 = 0.15$), and surface–atmosphere interaction ($w_5 = 0.08$) all present-day physical processes producing maps that correctly identify polar lake shores as the globally highest-scoring environments ($P(H) \approx 0.44$ at the present epoch; Kraken southern shore).

Prebiotic chemistry habitability. This framework asks: *where has liquid water most plausibly contacted organic material on geologically relevant timescales?* Neish et al. (2018) identified large impact craters as the clear answer. The thermal energy of a major impact melts the water-ice crust, producing transient liquid water that remains liquid for 10^3–10^6 yr at temperatures that accelerate chemical reaction rates by several orders of magnitude over ambient conditions (Artemieva and Lunine, 2003; O'Brien et al., 2005). Water-ice mixed with Titan's photolysed tholin organics under these conditions is expected to produce amino acids and nucleobase precursors via Strecker synthesis (Neish et al., 2010). Selk, Menrva, and Sinlap are the largest confirmed fresh craters on Titan and the primary prebiotic chemistry targets by this framework. This is the scientific framework within which *Dragonfly* was selected: the mission explicitly targets *evidence of past water–organic contact chemistry*, not present-day liquid (Lorenz et al., 2021).

Why the two frameworks give different site rankings. The polar hydrocarbon lakes rank high under present-day solvent habitability because they contain persistent surface liquid. They rank *lower* under the prebiotic chemistry framework precisely because liquid methane and ethane are non-polar: they do not dissolve the ionic species required for Strecker synthesis, and they never contact the water-ice crust in a chemically productive way. Conversely, impact craters rank low under present-day solvent habitability (no present liquid at the equatorial surface) but high under prebiotic chemistry habitability (confirmed past liquid water). Neish et al. (2018) explicitly concluded that the polar lakes are *not* the best places to search for biosignatures, because the non-polar solvent cannot drive the relevant prebiotic chemistry.

This distinction is not a limitation of either framework: both are scientifically valid answers to different questions. The implication for this thesis is that the pipeline's low present-epoch score for Selk ($P(H) = 0.24$) is not a contradiction of the *Dragonfly* mission's rationale — it is a *consequence* of targeting different habitability. Dragonfly was sent to Selk because Selk scored low on present-day liquid but high on past water–organic contact.

I.3 Quantitative Validation of Site Selection

The pipeline is a present-day solvent habitability model. By this metric, the north polar lake margins are the most habitable environments on Titan, reaching $P(H) \approx 0.44$ (Kraken southern shore) and 0.43 (Ligeia eastern margin). Selk scores $P(H) = 0.24$, below the prior mean of 0.33, because it lacks present-day surface liquid ($f_1 \approx 0.05$) and its methane-cycle and surface–atmosphere scores are equatorial-low. *This is scientifically correct*: a model that correctly captures present-day solvent habitability should rank polar lake shores above a dry equatorial crater.

The independent validation comes from the geomorphological diversity feature. Selk's $f_7 = 0.629$ is the highest value of any equatorial site and $7.8\times$ the global median of 0.081 — a result derived from Cassini SAR terrain data alone, without reference to mission planning. Geomorphological diversity captures the simultaneous presence of four terrain classes (dune, plains, labyrinth, crater floor) within 45 km, which is the Cassini-era observational proxy for the organic–water-ice mixing interface that defines the prebiotic chemistry target. The pipeline independently identifies Selk as the equatorial site with the highest terrain-class heterogeneity — precisely the physical attribute cited in the *Dragonfly* science rationale (Lorenz et al., 2021; Neish et al., 2018).

The polar lake margins represent a complementary, not competing, science target. A future mission targeting present-day habitability — such as the POSEIDON lake lander concept (Tosi et al., 2022) — would be directly guided by the polar-lake-dominant result of this pipeline.

I.4 Instrument–Feature Correspondence

DraMS atmospheric and surface composition measurements constrain f_3 (acetylene abundance) through depletion relative to photochemical model predictions, and f_2 (organic inventory) via amino acid and HCN polymer detection. DraGNS measures bulk hydrogen abundance, constraining f_8 (subsurface water proximity). The DraGMet EFIELD experiment directly measures the surface electric field, constraining f_4.

Enantiomeric excess in amino acids (expected >10% if biogenic vs. racemic for abiotic Strecker products) is the highest-priority biosignature for DraMS. Acetylene depletion below the photochemical steady-state prediction (McKay and Smith, 2005) would constitute strong but non-unique evidence for acetylenotrophy. Carbon isotopic fractionation (δ^{13}C depletion of ∼30–50 ‰ relative to atmospheric CH_4) would be consistent with biological methane cycling. All three biosignatures have plausible abiotic explanations and must be interpreted jointly.

I.5 Interpreting a Null Result

A null biosignature result at Selk would constrain specifically the "preserved impact-melt Strecker chemistry" pathway at that location and epoch. It would not constrain: (1) acetylenotrophy at the north polar lake margins, which score 15–20 percentage points higher than Selk in the present epoch and which *Dragonfly* does not visit; (2) life in the subsurface ocean, insensitive to surface sampling; or (3) extinct life from the LHB whose signatures may be buried below accessible depth. The habitability maps provide a rigorous framework for reporting a null result in terms of which specific pathways have been tested and which remain open.

J Comparison with Prior Habitability Frameworks

Several preceding studies have attempted to place planetary habitability on a quantitative footing. The *Earth Similarity Index* (Schulze-Makuch et al., 2011) scores planetary bodies against Earth on temperature, density, escape velocity, and surface chemistry, but produces a single scalar per body and cannot represent spatial or temporal variation. Hendrix et al. (2019) reviewed the habitability potential of icy ocean worlds using qualitative multi-criteria matrices, providing rich scientific context but no spatial resolution or probabilistic framework. Logistic regression and machine-learning habitability classifiers (e.g. Spiegel and Turner 2012) have been applied to exoplanet populations but require labelled training data that does not exist for Titan.

The Bayesian approach adopted in this thesis differs from all of these: it encodes scientific prior knowledge as probability distributions, updates them with observational data via conjugate updating, and returns credible intervals rather than point estimates — making uncertainty explicit rather than implicit. The choice of a Beta-conjugate (Beta-Binomial) model is motivated by the bounded $[0, 1]$ domain of habitability probability and by the analytical tractability that permits simultaneous evaluation at all 6.49 million grid pixels without numerical integration or MCMC. A Gaussian process or multivariate logistic regression could in principle capture feature correlations more richly, but would require either strong covariance priors or prohibitively dense labelled sampling to estimate the correlation structure from Cassini data alone. The Beta-conjugate approach is preferred as the most tractable framework that preserves physical interpretability at every pixel.

The closest methodological precedent is Affholder et al. (2021), who applied Bayesian statistical inference to Cassini INMS plume data from Enceladus, comparing the observed methane and hydrogen escape rates against models of abiotic serpentinisation and biotic methanogenesis. Their finding — that the observed methane escape rate is most parsimoniously explained by biological methanogenesis if the prior probability of life is sufficient — represents the most quantitatively rigorous Bayesian biosignature assessment yet published for a solar system body. The present work is directly inspired by this approach, extended from a single-point plume observation to a full-globe, multi-feature, multi-epoch spatial mapping.